



Unraveling the role of plant growth-promoting rhizobacteria in the alleviation of arsenic phytotoxicity: A review

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ABSTRACT

The toxic metalloid arsenic (As), is a major pollutant of soil and water, imposing severe health concerns on human lives. It enters the food chain mainly through As-contaminated crops. The uptake, translocation and accumulation of As in plant tissue are often controlled by certain soil-inhabiting microbial communities. Among them, indigenous, free-living As-resistant plant growth-promoting rhizobacteria (PGPR) plays a pivotal role in As-immobilization. Besides, the plant's inability to withstand As after a threshold level is actively managed by these PGPR increasing As-tolerance in host plants by a synergistic plant-microbe interaction. The dual functionality of As-resistant PGPR i.e., phyto-stimulation and minimization of As-induced phytotoxic damages are one of the main focal points of this review article. It is known that such PGPR having the functional arsenic-resistant genes (in *ars* operon) including As-transporters, As-transforming genes contributed to the As accumulation and detoxification/transformation respectively. Apart from assisting in nutrient acquisition and modulating phyto-hormone levels, As-resistant PGPR also influences the antioxidative defense system in plants by maneuvering multiple enzymatic and non-enzymatic antioxidants. Furthermore, they are effective in reducing membrane damage and electrolyte leakage in plant cells. As-induced photosynthetic damage is also found to be salvaged by As-resistant PGPR. Briefly, the eco-physiological, biochemical and molecular mechanisms of As-resistant PGPR are thus elaborated here with regard to the As-exposed crops.

1. Introduction

Arsenic (As) is 20th most ubiquitously found element in earth's crust, 14th in seawater, and 12th in the human body (Mandal and Suzuki, 2002). The terrestrial abundance of As ranges between 1.5–3 mg kg⁻¹ (Zhao et al., 2010). More than 200 As-containing mineral forms can be observed and approximately 60 % of them are arsenates [As(V)] and approximately 20 % of them are arsenites [As(III)]. The other forms include sulfides, sulfosalts, arsenides, oxides, silicates, and elemental As (Irshad et al., 2021; Mandal and Suzuki, 2002). Among numerous As ore minerals, the most common ones are arsenopyrite (FeAsS) and realgar (As₄S₄) (Mandal and Suzuki, 2002). In nature, As exists in both inorganic and organic forms with the inorganic species [As(III) and As(V)] being the most predominant ones (Shri et al., 2019). The highly oxidized inorganic form As(V) is more common in aerobic soils whereas the reduced form As(III) is customary for anaerobic soil environments

(Irshad et al., 2021; Tang and Zhao, 2020). Organic As species is least reported in soil and use of As-based agrochemicals and microbial methylation/demethylation of As are two main contributors of organic As to soil (Quaghebeur and Rengel, 2005). The major organic As species that are observed in soil includes monomethylarsonic acid (MMA), dimethylarsinic acid (DMA), and trimethylarsine oxide (TMAO) (Finnegan and Chen, 2012). Among others, arsenobetaine, arsenosugar, and methylated thioarsenates are noteworthy (Panda et al., 2010; Wang et al., 2020).

The sources of As contamination in the environment includes both natural sources (e.g., volcanic eruptions, weathering of rocks, hot spring discharges) as well as anthropogenic sources (e.g., usage of As-based agrochemicals, irrigation water, cattle waste; chemical, tannery, and paint industries, mining, smelting, sewage, municipal solid wastes, etc.) (Fig. 1). In soil, the concentration of As ranges from 0.1 to 40.0 mg kg⁻¹ depending upon geographic regions. While, in fresh water and seawater,

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the metalloid concentration lies between $0.15\text{--}0.45\ \mu\text{g L}^{-1}$ and $< 2\ \mu\text{g L}^{-1}$. The permissible limit of As in drinking water is $10\ \mu\text{g L}^{-1}$ as recommended by governing bodies such as the World Health Organization (WHO) and the United States Environmental Protection Agency (USEPA). The metalloid is usually found as a mixture of As(V) and As(III) in the air, where it ranges from 0.4 to $30\ \text{ng m}^{-3}$ (Hare et al., 2019). As a result, the human population gets exposed to this toxic metalloid directly or indirectly through the consumption of contaminated diet sources (Menon et al., 2020). Long-term exposure to this toxicant causes cardiovascular, peripheral vascular, neurological, gastrointestinal, hematological, renal, and respiratory diseases. As-induced toxicity is also marked by the development of skin lesions such as hyperkeratosis, hyperpigmentation, etc. (Fig. 1) (Abbas et al., 2018; Hare et al., 2019;

Menon et al., 2020; Rahman et al., 2020). Further, the toxic metalloid is considered as a group-1 human carcinogen by the International Agency for Research on Cancer (IARC) and is found to be causative of different types of cancer (Rahman et al., 2020; Ghosh et al., 2021).

The toxic effect As on organisms relies upon its form and redox status. The methylated As species were considered less toxic for a long time but recent developments suggest that some methylated As species are more toxic than their inorganic counterparts (Yan et al., 2019). The *modus operandi* of As(V) and As(III) in various organisms is well established. The toxic effect of As in plants is evident from its various physiological, biochemical, and molecular parameters. The most common physiological effects include leaf chlorosis, stunted growth, disruption of root architecture, and an overall reduction in growth and yield (Niazi

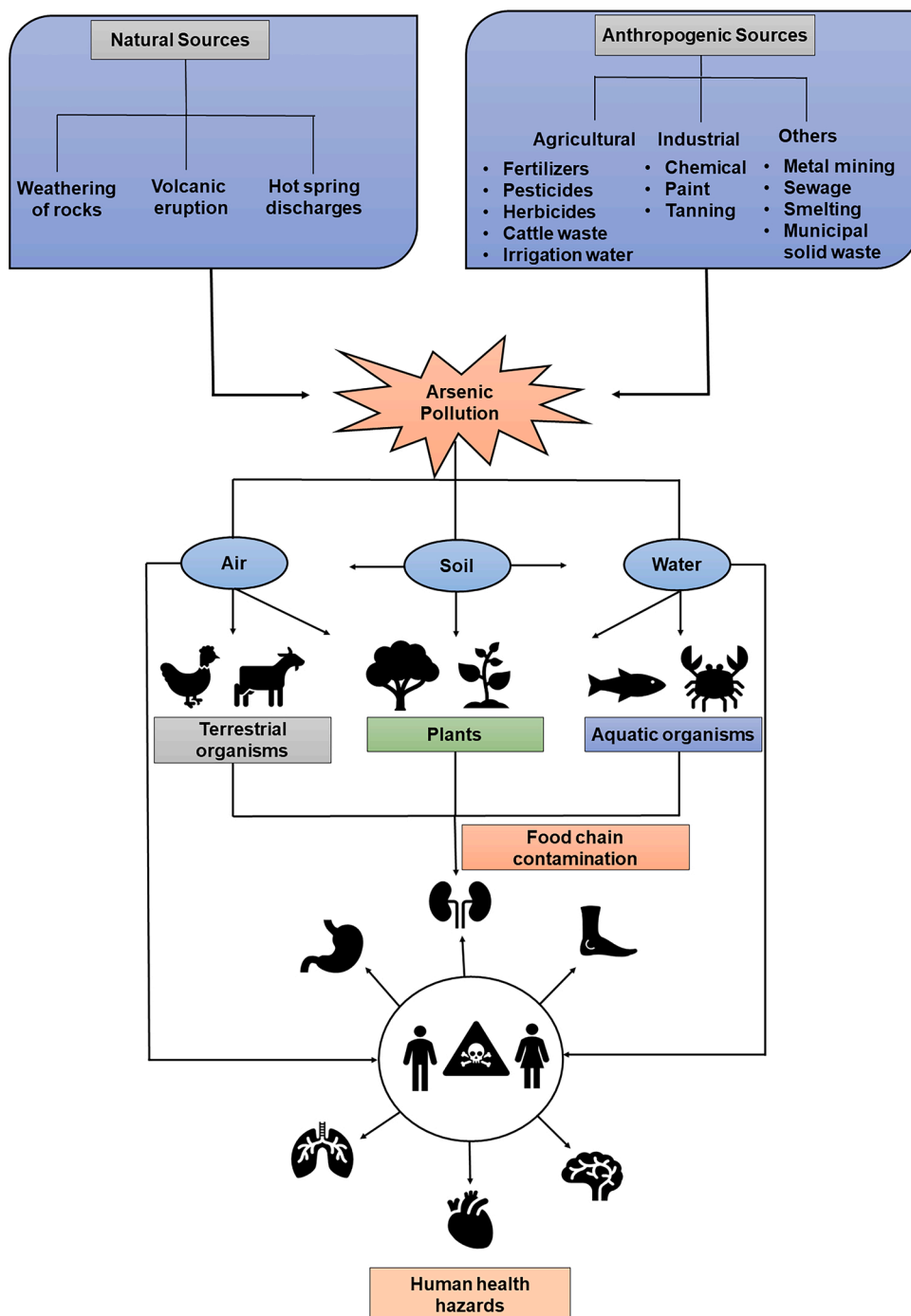


Fig. 1. Different sources of arsenic in the environment that subsequently contaminates the food chain causing severe health hazards to humans. The sources of arsenic include both natural (weathering of rocks, volcanic eruption, hot spring discharges, etc.) and anthropogenic (use of arsenic-based agrochemicals, contaminated irrigation water, cattle wastes; chemical, paint, tanning industry; smelting, mining, sewage, municipal solid wastes) sources. Arsenic enters the food chain via different means (air, soil, and water) and causes several human health problems (e.g., cardiovascular diseases, renal problems, respiratory diseases, neural problems, skin lesions, gastrointestinal diseases, and different types of cancers to name a few).

et al., 2017; Ronzan et al., 2019; Shri et al., 2009). By far, the most pronounced effect of the As(V) is the substitution of phosphate groups in various metabolic processes/products and the formation of unstable arsenical by-products.

On the other hand, As(III) tends to react with the sulfhydryl group, thereby affecting the protein metabolism of the plants (Abbas et al., 2018; Finnegan and Chen, 2012). The metalloid also disrupts nitrogen metabolism, photosynthesis, and membrane lipids. Moreover, As induces overproduction of reactive oxygen species (ROS) to dismantle the redox homeostasis of the cell (Abbas et al., 2018). To combat such adversaries, plants came up with numerous stress mitigating strategies that include As efflux, reduced As uptake, phytochelatin (PC), and/or glutathione arbitrated metalloid chelation followed by vacuolar sequestration, and antioxidant mediated detoxification. However, higher doses of the toxicant paired with other abiotic stress factors and differential response of plants to As-stress cumber these stress-responsive mechanisms.

From the past few decades, microorganisms were continuously exploited for their As-resistance properties. The As-detoxification processes in bacteria include– a. the *ars* operon mediated basic detoxification of inorganic arsenicals; b. As methylation/demethylation for detoxification of organic arsenicals; c. energy-yielding respiratory As (III) oxidation (*arr* system) and As(V) reduction (*aio/arx* system) (Yan et al., 2019).

Soil-dwelling microbes develop metal(loids) resistance property while residing in metal(loids) contaminated rhizosphere. The bacteria resistant to high concentrations of As and promote plant growth under the metalloid polluted soil concomitantly are referred to as As-resistant plant growth-promoting rhizobacteria (PGPR). This particular group of bacteria must have fulfilled some essential criteria like i) As-resistance property, ii) active root colonizing proficiency, and iii) presence of some plant growth-promoting (PGP) traits. The effect of these bacteria on plant growth under metalloid contaminated soil is multifaceted. They can reduce the metalloid uptake and its translocation to aerial parts in plants. These bacteria can sequester As within themselves reducing the toxic effects. Additionally, the PGP traits provide a much-needed impetus to plants to minimize the As-induced stress effects.

This phenomenon of PGPR-mediated bioremediation is widely viewed as an eco-friendly approach to minimize As-contamination in food crops (Irrshad et al., 2021). This review provides an in-depth analysis of As-resistance mechanisms in PGPR, the genetic basis of such mechanisms, and the multipurpose role of As-resistant PGPR in ameliorating As-induced phytotoxicity.

2. Arsenic resistance mechanisms in PGPR

Microorganisms capable of tolerating high concentrations of As possess numerous As-resistance mechanisms and can be exploited in bioremediation methods. Bacterial As-resistance mechanisms are broadly classified into intra- or extracellular sequestration and active extrusion of the metalloid. Besides, the enzymatic transformation of As is also accomplished via bacterial enzymes.

Biosorption and bioaccumulation are two well-known methods that enable As sequestration extracellularly and intracellularly respectively. Bacterial cells are often regarded as an excellent bio-sorbent due to their high surface-to-volume ratio and the presence of numerous active chemisorption sites (Sharma and Archana, 2016). Biosorption is an energy-independent physicochemical process that scavenges heavy metal(loids) onto the outer surface of the bacterial cell (Pratish et al., 2018; Sher and Rehman, 2019). As the toxicants cannot enter the cell it does not interfere with the bacterial metabolism and does not exert any toxic effects (Sher and Rehman, 2019). Peptidoglycan and phosphate groups of gram-positive and gram-negative bacteria can serve as cationic and anionic binding sites respectively (Pratish et al., 2018). The extracellular polymeric substances (EPS), mostly a complex mixture of polysaccharides, mucopolysaccharides, proteins, lipids, humic

substances, and uronic acids are a high molecular weight microbial secretory by-product and can also function as an absorbent of the toxic metal(loids) (Mukherjee et al., 2019; Pratish et al., 2018; Sharma and Archana, 2016; Slyemi and Bonnefoy, 2012). The abundance of components like acetamido, amine, sulfhydryl, and carboxyl groups in proteins and phosphodiester, phosphate, and hydroxyl groups in polysaccharides disseminate an overall negative charge to EPS that further favors metal(loids) sequestration (Mukherjee et al., 2019). Sulfate-reducers and iron-oxidizers are also capable of precipitating the metalloid outside the cell using the end products of energy metabolism such as H_2S and Fe^{3+} (Slyemi and Bonnefoy, 2012). On the other hand, bioaccumulation is an energy-dependent, slow and irreversible process whereby heavy metal(loids) are accumulated within the bacterial cell after being channelled through ion pumps, ion channels, endocytosis, and lipid permeation (Pratish et al., 2018; Sher and Rehman, 2019). Contrary to biosorption, this is a metabolism-dependent process and exerts certain toxic effects on bacteria (Sher and Rehman, 2019). The metal(loids) is probably scavenged via cysteine-rich metallothioneins inside the cell (Sher and Rehman, 2019; Tabak et al., 2005). The process of biosorption and bioaccumulation is widely distributed in PGPR isolated from As-contaminated fields. These are viewed as shielding mechanisms to withstand metalloid toxicity in the contaminated environment.

Moreover, biofilm formation among these PGPR is also reported. The biofilm matrix is highly structured, usually composed of EPS, extracellular nucleic acids, and proteins. It encloses cell communities along its surface that further restrict the metalloid diffusion (Armendariz et al., 2015; Vezza et al., 2020). The active extrusion of the metalloid from the bacterial cell is by far the most studied mechanism. The *ars* operon (*ArsB/ArsAB*) mediated extrusion is widely distributed and well characterized. Another family of As(III) efflux permease, *Acr3*, is found in every domain of life. It is a uniporter and selectively expels As(III) from bacterial cells using proton motive force (Fu et al., 2009). *Sinorhizobium meliloti* utilizes a unique detoxification pathway where As(III) is extruded via an aquaglyceroporin, encoded by *AqpS*. Surprisingly, *AqpS* is encoded within *ars* operon replacing *arsB* (Yang et al., 2005). Arsenic detoxification via *ars* operon along with the other major bacterial As resistance mechanism i.e., biotransformation of As which include As(V) reduction, As(III) oxidation, and As methylation/demethylation, is discussed in detail in the following section.

3. Genetic basis of arsenic resistance in PGPR

3.1. Arsenite and arsenate uptake systems

Specific transporter for As is yet to be reported in microorganisms probably due to the acute toxicity of the metalloid. Instead, the metalloid is taken up by the microbes through various non-selective membrane transporters meant for other molecules (Slyemi and Bonnefoy, 2012). As(V), the pentavalent form, is chemically analogous to phosphate and is taken up through phosphate transport systems. Two phosphate transporters– phosphate inorganic transport (Pit) and phosphate specific transport (Pst) are particularly involved in this process (Fig. 2). Among them, Pit is a non-specific transport system and transports phosphate more efficiently.

Meanwhile, Pst being a specific transporter, transports phosphate with high affinity but with lower efficiency (Rosenberg et al., 1977; Willsky and Malamy, 1980a, b). On the other hand, As(III) possesses structural similarity with glycerol and is taken up via aquaglyceroporin channel GlpF, a member of the major intrinsic protein (MIP) family (Meng et al., 2004; Tsai et al., 2009). However, disruption of *glpF* gene in *E. coli* does not result in complete resistance to As(III) and hence the existence of other As(III) uptake pathways are not improbable (Meng et al., 2004; Sanders et al., 1997). Hexose transporters are also eventually found to be involved in As(III) uptake in higher organisms (Tsai et al., 2009).

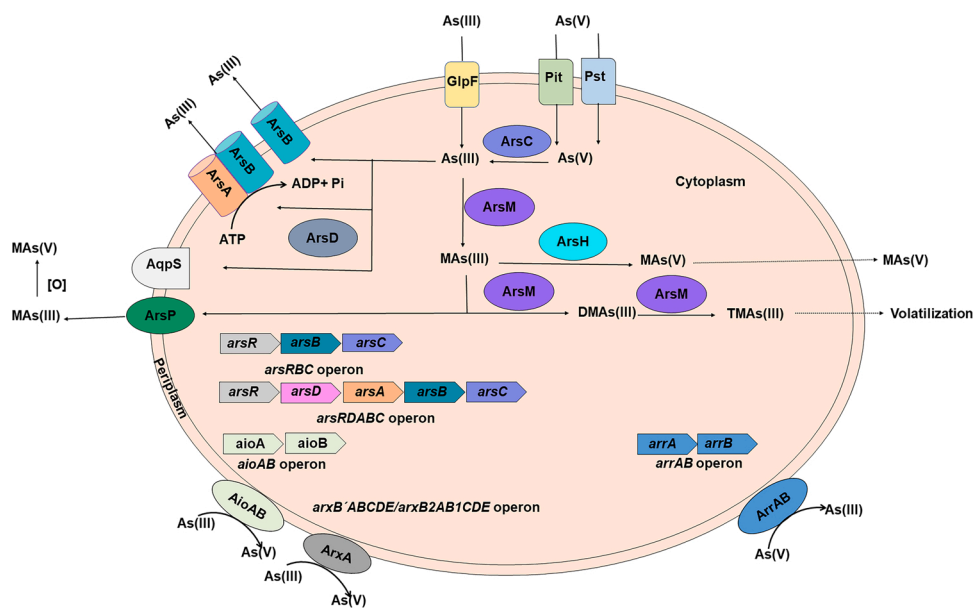


Fig. 2. An overview of arsenic resistance mechanisms in bacteria. Arsenate [As(V)] being a phosphate analog enters through phosphate inorganic transport (Pit) and phosphate specific transport (Pst), whereas arsenite [As(III)] enters via aquaglyceroporin channel GlpF. Once inside the cell, As(V) is reduced to As(III) via cytoplasmic As(V) reductase (ArsC). The reduced As(III) as well as the As(III) directly taken through GlpF extruded out of the cell for detoxification by an As(III)-efflux permease, ArsB or by ArsAB (ArsA, an ATPase coupled with ArsB) or by an aquaglyceroporin AqpS. The Ars (ArsA, B, and C) proteins involved in arsenic detoxification are encoded either by *arsRBC* operon or *arsRDABC* operon. ArsR and ArsD proteins are regulatory elements. The As(V) is also reduced via ArrAB system (periplasmic respiratory arsenate reductase) encoded by *arr* operon. On the other hand, As(III) is converted to less toxic As(V) by respiratory As(III)-oxidase encoded by *aio/arr* operon. Arsenite can be methylated into methylarsenite [MAs(III)], dimethylarsenite [DMAs(III)], and trimethylarsenite [TMAs(III)] by ArsM. The end product TMAs(III) can be volatilized from the cell. The further detoxification of MAs(III) can be achieved through a MAs(III) efflux permease ArsP. Moreover, these trivalent

organoarsenicals can be oxidized to less-toxic pentavalent organoarsenicals. For example, MAs(III) are converted to MAs(V) via ArsH.

3.2. Arsenic detoxification system involving *ars* operon

The most extensively studied pathway for As detoxification in microbes involves *ars* operon. The operon is either chromosome-borne or plasmid-borne and is present in gram-positive as well as in gram-negative bacteria (Tsai et al., 2009). The organization of *ars* operon is relatively diverse and is composed of different gene clusters in different microbial strains (Table 1). The most common among them is the three gene operon –*arsRBC* (e.g., *E. coli* genome and *Staphylococcus aureus* plasmid pI258) and five gene operon –*arsRDABC* (e.g., *E. coli* plasmid R773) (Carlin et al., 1995; Chen et al., 1986; Ji and Silver, 1992) (Fig. 2).

The basic detoxification through the As resistance system is more or less similar in all bacteria and confers resistance to both As(V) and As(III). It is a two-step process in which As(V) is reduced to As(III) with the help of a cytoplasmic As(V) reductase after being taken up through phosphate transporters (Pit and Pst). The reduced As(III) [as well as the As(III) taken up directly through GlpF] is extruded out of the cell for further detoxification (Fig. 2).

ArsC is a small, monomeric, and cytoplasmic As(V) reductase that reduces As(V) into As(III) through a three cysteine thiol nucleophilic attacking cascade (Slyemi and Bonnefoy, 2012). In prokaryotes, two families of As(V) reductase have been reported, which differ mainly in their structures, electron donors for reducing mechanisms, and location of the catalytic cysteine residues. The thioredoxin-coupled ArsC reductase, represented by *Staphylococcus aureus* plasmid pI258, uses thioredoxin as an electron source (Ji and Silver, 1992; Mukhopadhyay and Rosen, 2002). Besides, thioredoxin reductase and NADPH are also required to complete the catalytic event (Zegers et al., 2001). The second family, i.e., glutaredoxin-coupled ArsC typified by *E. coli* R773 plasmid *ars* locus, utilizes glutaredoxin as a reducing agent. The reaction also requires reduced glutathione (GSH) (Shi et al., 1999). The ArsC of *E. coli* R773 plasmid is structurally unrelated with *Staphylococcus aureus* plasmid pI258 (shares only 12 % similarity in tertiary structure) which shows more resemblance with phosphatases (Zegers et al., 2001).

It has already been mentioned that the basic detoxification of As(III) takes place through extrusion from the cell. ArsB is an example of such efflux permease that primarily uses membrane potential to expel As(III) out of the cell (Dey and Rosen, 1995). On the other hand, an As(III)

induced ATPase, ArsA can also serve as an efflux protein (Kuroda et al., 1997; Rosen et al., 1990). However, a higher level of resistance is reported when these two (ArsAB) forms a tight membrane-bound complex of ATP driven As(III) efflux pump, in which ArsA serve as a catalytic subunit (periplasm-located) and ArsB serves the purpose of a membrane anchor for the catalytic subunit (Tisa and Rosen, 1990).

The two other main components of the *ars* operon i.e., ArsR and ArsD are generally regulatory elements. The ArsR is an As(III)-responsive trans-acting repressor protein that negatively regulates other genes of the *ars* operon (Wu and Rosen, 1991). It belongs to a family of small metalloregulatory proteins (the family includes ArsR, CadC, and SmtB) and possesses three distinct domains – a metal-binding domain, a DNA-binding domain, and a dimerization domain (Wu and Rosen, 1991; Xu et al., 1998; Yan et al., 2019). Recently, a novel methylarsenite [MAs(III)]-responsive ArsR from *Shewanella putrefaciens* has also been reported (Chen et al., 2017). Initially, ArsD was considered as a weak transcriptional repressor that binds to the same promoter site as ArsR. However, ArsR and ArsD do not have any sequence similarity (Chen and Rosen, 1997; Wu and Rosen, 1993). These two together were thought to control the basal and maximal expression levels of the *ars* operon, respectively (Xu et al., 1998). More recent studies revealed the specific role of ArsD, which function as a metallochaperone and transfers As(III) to the metal-binding site of ArsA efflux permease (Lin et al., 2006; Yang et al., 2010). This interaction increases the ATPase activity of ArsA even at the lowest As(III) concentrations, thereby increasing As(III) extruding efficacy of the bacteria (Lin et al., 2007).

One striking feature of this whole detoxification mechanism is that the less toxic As(V) is converted into more toxic As(III) to extrude them using the pre-existing As(III) efflux pumps. As(V) specific efflux pumps are yet to be reported in microbes. The roots of this particular phenomenon can be drawn back to the early days of evolution when the earth's atmosphere was oxidizing and hence As was found mostly in its trivalent form. Therefore, organisms have developed As(III) detoxification mechanism and not As(V). Later, when the atmosphere becomes oxidizing, organisms learned to use pre-existing detoxification machinery to counter As(V). It is also speculated that during this period As(V) reductases might have been evolved from phosphatases to fine-tune the prevailing detoxification system (Rosen, 2002; Slyemi and Bonnefoy,

Table 1

Key genes involved in uptake and resistance mechanisms of arsenic in bacteria.

Genes	Nature of the encoded product	Specific function(s)	References
<i>aioA</i>	Large catalytic subunit of As(III) oxidase (AioAB) contains a molybdenum center and a [3Fe-4S] cluster	As(III) oxidation	Muller et al., 2003; Santini and vanden Hoven, 2004
<i>aioB</i>	Small subunit of As(III) oxidase (AioAB) shows similarity to Rieske [2Fe-2S] protein domains	As(III) oxidation	Muller et al., 2003; Santini and vanden Hoven, 2004
<i>aqpS</i>	Aquaglyceroporin	Facilitates efflux of reduced As(III)	Yang et al., 2005
<i>arrA</i>	Large subunit of respiratory arsenate reductase (Arr)	Respiratory As(V) reduction	Krafft and Macy, 1998; Kruger et al., 2013
<i>arrB</i>	Small subunit of respiratory arsenate reductase (Arr)	Respiratory As(V) reduction	Krafft and Macy, 1998; Kruger et al., 2013
<i>arrC</i>	Arsenate reductase membranous subunit	Probably membrane anchoring, although specific function not known yet	Richey et al., 2009; van Lis et al., 2013
<i>arrD</i>	Arsenate reductase chaperone	Act as a chaperone	Van Lis et al., 2013
<i>arsA</i>	Oxyanion stimulated ATPase located on periplasm	Acts as a catalytic subunit of ArsAB As(III) efflux pump Extrudes As(III) utilizing membrane potential	Tisa and Rosen, 1990
<i>arsB</i>	Transmembrane As(III) efflux pump	Serve as an anchor in the inner membrane when present together with ArsA to form an efficient ATP-driven As(III) efflux pump	Dey and Rosen, 1995; Tisa and Rosen, 1990
<i>arsC</i>	Cytoplasmic arsenate reductase	Converts intracellular As(V) to As(III) before extruding from the cell using glutaredoxin or thioredoxin as an electron source	Ji and Silver, 1992; Yan et al., 2019
<i>arsD</i>	Trans-acting repressor protein/ As(III) metallochaperone	Controls maximal expression levels of the <i>ars</i> operon/ Transfers As(III) to the metal-binding site of ArsA of ArsAB efflux pump	Lin et al., 2006, 2007; Wu and Rosen, 1993; Xu et al., 1998; Yang et al., 2010
<i>arsH</i>	NADPH-flavin mononucleotide (FMN)- dependent oxidoreductase	Oxidizes trivalent methylarsenicals into their pentavalent form	Chen et al., 2015a
<i>arsM</i>	As(III)-S-adenosylmethionine methyltransferase	Transfers methyl groups from S-adenosylmethionine (AdoMet) to As(III)	Qin et al., 2006
<i>arsP</i>	Methylarsenite efflux permease	Detoxification of MAs(III)	Chen et al., 2015b; Shen et al., 2014
<i>arsR</i>	Trans-acting transcriptional repressor	Negative regulator of other genes of <i>ars</i> operon. Controls basal expression levels of <i>ars</i> operon	Wu and Rosen, 1991; Xu et al., 1998
<i>arxA</i>	Arsenite oxidase within DMSO reductase family of molybdenum oxidoreductases	Encodes molybdenum containing arsenite oxidase alpha subunit	Zargar et al., 2010, 2012
<i>arxB</i>	Three [4Fe-4S] cluster containing protein	Encodes [4Fe-4S] containing protein	Zargar et al., 2010, 2012
<i>arxB' / arxB2</i>	Another [4Fe-4S] cluster containing protein	Encodes [4Fe-4S] containing protein	Zargar et al., 2010, 2012
<i>arxC</i>	Quinol-oxidizing membrane protein/ homolog of NfrD, a transmembrane protein involved in redox interactions with quinones	Encodes for putative regulatory proteins	Zargar et al., 2010, 2012
<i>arxD</i>	TorD-like protein, a cytoplasmic chaperone	Protects molybdoenzymes from degradation until insertion of molybdenum cofactor and also in molybdoenzyme biosynthesis	Zargar et al., 2010, 2012
<i>arxE</i>	Resembles peptidylprolyl isomerase	Aids in protein folding	Zargar et al., 2010
<i>glpF</i>	Aquaglyceroporin	As(III) uptake	Meng et al., 2004
<i>pit</i>	Non-specific phosphate transporter which transports phosphate more efficiently	As(V) uptake	Rosenberg et al., 1977; Willsky and Malamy, 1980a, b
<i>pst</i>	Specific phosphate transporter and transports phosphate with high affinity	As(V) uptake	Rosenberg et al., 1977; Willsky and Malamy, 1980a, b

As(III)- arsenite; As(V)- arsenate.

2012).

The As resistance mechanisms in As-resistant PGPR are diverse and are not well documented. Fluorescence *in situ* hybridization analysis of the rhizobacterial community of *Cirsium arvense* (L.) grown in an As-contaminated soil revealed that α -, β -, and γ -Proteobacteria are the predominant phylogenetic bacterial groups. The study also stated that the As-resistant isolates mostly belonged to *Bacillus*, *Achromobacter*, *Brevundimonas*, *Microbacterium*, and *Ochrobactrum*. Multiple strains were found to reduce As(V) and oxidize As(III) and one of them was able to do both simultaneously. Most of these strains possessed either *arsC* [encodes As(V) reductase] or *arsB* and/or *acr3* [encodes As(III) efflux pumps] (Cavalca et al., 2010). The presence of *arsB* and *acr3* genes have been further reported from isolates of Firmicutes and γ -Proteobacteria group encompassing several bacterial genera such as *Bacillus*, *Luteimonas*, *Escherichia*, *Psychrobacter*, *Klebsiella*, *Citrobacter*, *Acinetobacter*, and *Stenotrophomonas* (Qamar et al., 2017). The presence of *arsC* gene is also reported from *Staphylococcus arlettae* isolated from the rhizosphere of *Brassica juncea* (Srivastava et al., 2013). In *Bacillus aryabhattai*, multiple *ars* operon genes viz *arsAB*, *arsD*, *arsR*, *arsH*, *arsC*, *arsA*, and *arsB* were detected from its chromosomal DNA (Singh et al., 2016). The strain was later identified as a potent As-resistant PGPR (Ghosh et al., 2018). High-throughput sequencing of *Azospirillum brasilense* Cd plasmid

revealed a genetic organization similar to *ars* operon. The module encompasses *arsR1*, *arsR2*, *arsC*, *acr3*, and *arsH* (Veza et al., 2020).

3.3. Biotransformation of arsenic

3.3.1. Reduction of arsenate

Two types of As(V) reduction systems are being operated in bacteria – the cytoplasmic As(V) reduction system mediated by ArsC and the periplasmic respiratory As(V) reduction system mediated by Arr proteins (Kruger et al., 2013). The periplasmic respiratory As(V) reduction system is often referred to as dissimilatory As(V) reduction and is mediated by several obligate and facultative anaerobes that utilize As(V) as a terminal electron acceptor. These microbes are also known as dissimilatory As(V)-reducing prokaryotes (DARPs). The DARPs can utilize a wide range of electron donors for subsequent As(V) reduction such as inorganic hydrogen and sulfide, organic acids (acetate, lactate, formate, pyruvate, etc.), sugars, ethanol, and complex aromatic compounds like benzoate, toluene, and phenol (Kruger et al., 2013; Slyemi and Bonnefoy, 2012). Bacteria from diverse phylogenetic groups comprise DARPs – Gram-positive Eubacteria, β -, γ -, ϵ - Proteobacteria, thermophilic Eubacteria, and Crenarchaeota (Oremland and Stolz, 2003; Pérez-Espino et al., 2009; Slyemi and Bonnefoy, 2012). Examples

of As(V) reducing strain includes strain MIT-13, SES-3, OREX-4 (*Desulfotomaculum auripigmentum*), *Alkaliphilus oremlandii* sp. nov. strain OhLAs, *Chrysiogenes arsenates*, *Bacillus selenitireducens* strain MLS10, and *Shewanella* strain ANA-3, etc. (Afkar et al., 2003; Ahmann et al., 1994; Fisher et al., 2008; Krafft and Macy, 1998; Laverman et al., 1995; Malasarn et al., 2008; Newman et al., 1997; Saltikov and Newman, 2003). The respiratory reduction of As(V) in *Chrysiogenes arsenates* was found to be catalyzed by a respiratory As(V) reductase (Arr) which supposedly is a heterodimer ($\alpha_1\beta_1$) composed of two subunits ArrA (87 kDa) and ArrB (29 kDa). The heterodimer complex contains molybdenum, Fe, acid-labile sulfur, and Zn as co-factors. The N-terminus of ArrA is reflective of many prokaryotic molybdenum-containing polypeptides while the N-terminus of ArrB is similar to that of Fe-S proteins. The enzyme complex is distinctly different from ArsC [non-respiratory As(V) reductase] of *E. coli* and *Staphylococcus aureus* (Krafft and Macy, 1998).

The Arr protein complex purified from *Bacillus selenitireducens* strain MLS10 also has a heterodimeric configuration of ArrA (110 kDa) and ArrB (34 kDa) (Afkar et al., 2003). In *Alkalilimnicola ehrlichii*, the *arr* operon was found to contain additional genes (*arrC* and *arrD*) that encode accessory proteins such as ArrC and ArrD. ArrC is a membrane anchoring subunit that shares homology with NfrD and ArrD is similar to a chaperon TorD (Richey et al., 2009). Additional genes in *arr* operon have also been reported from several other bacteria such as *Desulfotomaculum hafniense*, *Alkaliphilus oremlandii*, *Bacillus selenitireducens*, and strain MLMS-1 (van Lis et al., 2013).

However, in PGPR only the phenomenon of cytoplasmic As(V) reduction followed by active extrusion is reported to date (Fig. 2). The examples of such PGPR along with responsible genes are already discussed in the previous section.

3.3.2. Oxidation of arsenite

Microbially-influenced oxidation of As(III) to a less toxic As(V) is of immense importance in bioremediation of As. For a long period, the oxidation of As(III) was thought to be a mere detoxification process. Later, it was found that microbes can also harness energy from the process. A wide spectrum of phylogenetically and ecologically different bacteria are found to oxidize As(III) which includes heterotrophs, mixotrophs, and chemolithoautotrophs (Slyemi and Bonnefoy, 2012). Some authors tend to categorize As(III)-oxidizing bacteria into two classes based on their energy source– i) heterotrophic As(III) oxidizers (HAOs) and ii) chemolithoautotrophic As(III) oxidizers (CAOs) (Oremland and Stolz, 2003; Mitra et al., 2017). The HAOs may not always derive energy from As(III) oxidation. They generally prefer organic carbon as an energy source to oxidize As(III). However, HAOs like *Hydrogenophaga* NT-14 oxidizes As(III) to yield energy in the presence of oxygen (vanden Hoven and Santini, 2004). On the flip side, As(III) serves as an electron donor and itself is converted into As(V), while CO₂ act as a carbon source, and either oxygen (as in *Rhizobium* strain NT-26 and *Thiomonas arsenivorans*) or nitrate (as in *Alkalilimnicola ehrlichii* MLHE1, *Acidovorax* strain NO1) functions as an ultimate electron acceptor (Bryan et al., 2009; Huang et al., 2012; Oremland et al., 2002; Santini et al., 2000). In *Ectothiorhodospira* sp. strain PHS-1, As(III) is utilized as an electron donor for anoxygenic photosynthesis (Kulp et al., 2008).

In most As(III) oxidizing microorganisms, the reaction is catalyzed by an enzyme complex known as AioAB. This enzyme has been well characterized in *Alcaligenes faecalis* and a β -Proteobacterial strain ULPAs1. In *Alcaligenes faecalis*, the enzyme resides as a 100 kDa heterodimer ($\alpha_1\beta_1$). It consists of a molybdopterin cofactor and a unique high potential [3Fe-4S] cluster (similar to DMSO reductase family) in a large subunit (AioA-88 kDa) instead of a more common [4Fe-4S] cluster and a Rieske type cluster [2Fe-2S] in a small subunit (AioB- 14 kDa) (Anderson et al., 1992; Ellis et al., 2001; Muller et al., 2003). The AioAB complex (219 kDa) is also characterized in a chemolithoautotrophic As(III) oxidizer NT-26 in which it is periplasmic with a large subunit AioA (98 kDa) and a small subunit AioB (14 kDa) with heterotetrameric configuration ($\alpha_2\beta_2$). The large subunit (AioA) contains a molybdenum center

and [3Fe-4S] cluster just like other members of DMSO reductase family; meanwhile, the small subunit (AioB) contains Rieske centers [2Fe-2S] of cytochrome bc₁ and b₆f complexes (Santini and vanden Hoven, 2004). The similar architecture of As(III) oxidase, i.e. [3Fe-4S] cluster and Rieske center like the cluster in large and small subunit respectively, has also been reported in *Arthrobacter* sp. 15b (Prasad et al., 2009).

Another As(III) oxidase, ArxA also found responsible for As(III) oxidation (coupled with nitrate reduction for chemolithoautotrophic growth) was initially detected in *Alkalilimnicola ehrlichii* strain MLHE-1 (Zargar et al., 2010). The ArxA, a component of *arx* operon, is a bifunctional enzyme having both As(V) reduction (*in vitro*) and As(III) oxidation (*in vivo* as well as *in vitro*) capability. It belongs to the DMSO reductase family of molybdenum oxidoreductase family having a greater evolutionary affinity towards ArrA rather than AioA (Zargar et al., 2010, 2012). Apart from *arxA*, the operon also comprised of several other genes such as *arxB*, *arxC*, *arxD*, and *arxE* (Table 1) (Zargar et al., 2010).

The *aio/arx* operon regulates the respiratory oxidation of As(III) in particular which is a rarity in PGPR (Fig. 2). In a rare instance, the presence of *aoxB* (*aioB*) has been detected in a PGPR, *Bacillus flexus* (Das et al., 2016). Many PGPR are supposedly capable of oxidizing As(III) but the underlying genetic mechanism is yet to be uncovered. These As(III)-oxidizing PGPR could have other genes that may elucidate the basis of their As(III)-oxidizing ability.

3.3.3. Methylation/demethylation

Arsenic methylation/demethylation catalyzed by bacterial enzymes is a major biotransformation process that further promotes As detoxification (Fig. 2). Methylation of As species followed by subsequent volatilization was once considered to be a major detoxification process. However, recent studies revealed that some methylated As species produced during this procedure are relatively more toxic than the inorganic ones (Slyemi and Bonnefoy, 2012). Probably due to this reason they do not tend to accumulate within bacterial cells and subsequently detoxified using various pathways. Moreover, the volatile arsenicals can be reverted to non-volatile compounds in the environment by UV radiation (Mestrot et al., 2011). The methylation of As(III) is mainly catalyzed by As(III) S-adenosylmethionine methyltransferase (ArsM), an enzyme encoded by *arsM* gene. The ArsM transfers methyl groups from S-adenosylmethionine to As(III). This methylation of As(III) is a multistep process leading to the formation of trimethylarsenite [TMAs(III)] with intermediates like MAs(III) and DMAs(III). The terminal product TMAs(III) is regarded as a non-toxic compound and its volatilization can be marked as a detoxification method (Qin et al., 2006; Yuan et al., 2008). The trivalent methylarsenicals can be oxidized to pentavalent ones [MAs(V), DMAs(V), TMAsO(V)] under aerobic conditions. For instance, ArsH catalyzed oxidation of MAs(III) to MAs(V). ArsH, which was initially anticipated as a reductase found to be an NADPH-flavin mononucleotide (FMN)-dependent oxidoreductase that catalyzes mixed oxidation-reduction reaction (Yan et al., 2019). The detoxification of MAs(III) is also achieved through efficient efflux through a MAs(III) efflux permease ArsP (Chen et al., 2015b) (Fig. 2).

It also confers resistance to other trivalent organoarsenicals like roxarsone [Rox(III)] and phenylarsenite [PhAs(III)] (Chen et al., 2015b; Shen et al., 2014). The prevalence of *arsH* has been detected in PGPR as discussed in Section 3.2. Its gene product ArsH usually mediates oxidation of trivalent organoarsenicals as demonstrated in *Sinorhizobium meliloti* and *Pseudomonas putida* and no such role of ArsH for inorganic As was reported in those bacteria. However, the involvement of ArsH in inorganic As metabolism cannot be overruled as mutations in *arsH* affected the As(V) and As(III) resistance in *Yersinia enterocolitica* and *Serratia marcescens* (Chen et al., 2015a; Yang and Rosen, 2016; Vezza et al., 2020). The presence of *arsM* gene in a *Streptomyces* strain isolated from the As-contaminated rice rhizosphere is also detected. However, the gene seemed to be phylogenetically different from *arsM* gene present in other bacteria. Heterologous expression of this *Streptomyces arsM*

gene in *E. coli* was found to convert As(III) into organoarsenicals such as DMA and TMAO (Kuramata et al., 2015).

4. Multidimensional role of arsenic-resistant PGPR in alleviating arsenic phytotoxicity

4.1. Assistance in nutrient acquisition

4.1.1. Nitrogen

The nitrogen-fixing trait of PGPR help plants to overcome their inability to fix atmospheric nitrogen (N_2). The biochemical process of nitrogen fixation involves reducing N_2 into ammonia (NH_3) with the help of an enzyme complex known as nitrogenase. The process of nitrogen fixation operates symbiotically (in legumes and non-legumes) and non-symbiotically (through free-living N_2 fixers). Arsenic is long known for its interference in the symbiotic N_2 fixation mechanism and the nitrogen assimilation mechanism (Bianucci et al., 2018; Finnegan and Chen, 2012). Arsenic caused a marked decline in nitrogen assimilatory enzymes such as nitrate reductase, nitrite reductase, glutamine synthetase, and glutamate synthase (Ghosh et al., 2013; Jha and Dubey, 2004). The decline of the nitrogen assimilatory enzymes is accompanied by decreased affinity towards their substrates (Jha and Dubey, 2004). Total soluble nitrogen content and free amino acid contents are also adversely affected by As (Ghosh et al., 2013). Arsenic-resistant PGPR might be helpful under such a scenario performing a pivotal role in As bioremediation as well as in soil nitrogen improvement thereby improving the plant's overall growth. Several studies have demonstrated the effect of As on the N_2 fixation mechanism of the legume-rhizobium system. In most of the cases, As-affected plants such as *Medicago sativa*, *Vigna mungo*, and *Glycine max* showed severe symptoms like inhibition of root growth, necrosis, root hair damage, decrease in root hair numbers, reduction in nodule numbers, and decreased N_2 fixing zones (Mandal et al., 2011; Pajuelo et al., 2008; Reichman, 2007).

Besides, As seems to affect nitrogenase activity, nitrogen content, nodule redox state, and overall plant growth (Armendariz et al., 2019; Bianucci et al., 2018; Mandal et al., 2011). It was also revealed that early steps in nodulation are mostly affected (Bianucci et al., 2018; Pajuelo et al., 2008) (Fig. 3). A significant decrease in early nodulin genes correlates with As-induced reduction of nodulation (Lafuente et al., 2010). However, using appropriate As-resistant strain such as *Sinorhizobium* sp. strain MA11 helps to re-establish symbiotic interaction in alfalfa under moderate contamination, although that cannot prevent early senescence of the nodule (Pajuelo et al., 2008). Similar results were obtained using a *Rhizobium* strain VMA301 in *Vigna* and *Bradyrhizobium* strains in soybean (Bianucci et al., 2018; Mandal et al., 2011). To get rid of As toxicity plants are biased towards establishing symbiotic interaction as evidenced from enhanced chalcone synthesis which is involved in the first step of legume-rhizobia crosstalk (Lafuente et al., 2015). Not only the symbiotic N_2 fixers but also their associative symbiotic and free-living counterparts prove to be beneficial for plants under As contamination. Among associative symbiotic N_2 fixers, *Azospirillum brasilense* Cd was found to fix N_2 efficiently at relatively high As(V) (500 μ M) and As(III) (25 μ M) levels *in vitro* (Veza et al., 2020). Double inoculation with *Bradyrhizobium japonicum* ER109 and *Azospirillum brasilense* Az39 proved effective in As-stressed soybean (Armendariz et al., 2019). Shagol et al. (2014) isolated at least 13 potent As-resistant PGPR from As-contaminated sites mostly belongs to the genera of *Pseudomonas* and *Rahnella* with N_2 fixing abilities. *Bacillus aryabhattai* AS6 and *Pantoea dispersa* are other examples of bacterial strain that fixes N_2 (approx. 0.32 and 0.34 μ g N fixed/hour/mg protein respectively) under high As concentrations. These strains are also found to promote the growth of rice seedlings alleviating As phytotoxicity (Ghosh et al., 2018, 2021). Taken all this data together, we can safely say that N_2 fixing and As-resistant PGPR should be further exploited to improve soil nitrogen content and eliminate phytotoxic effects of As.

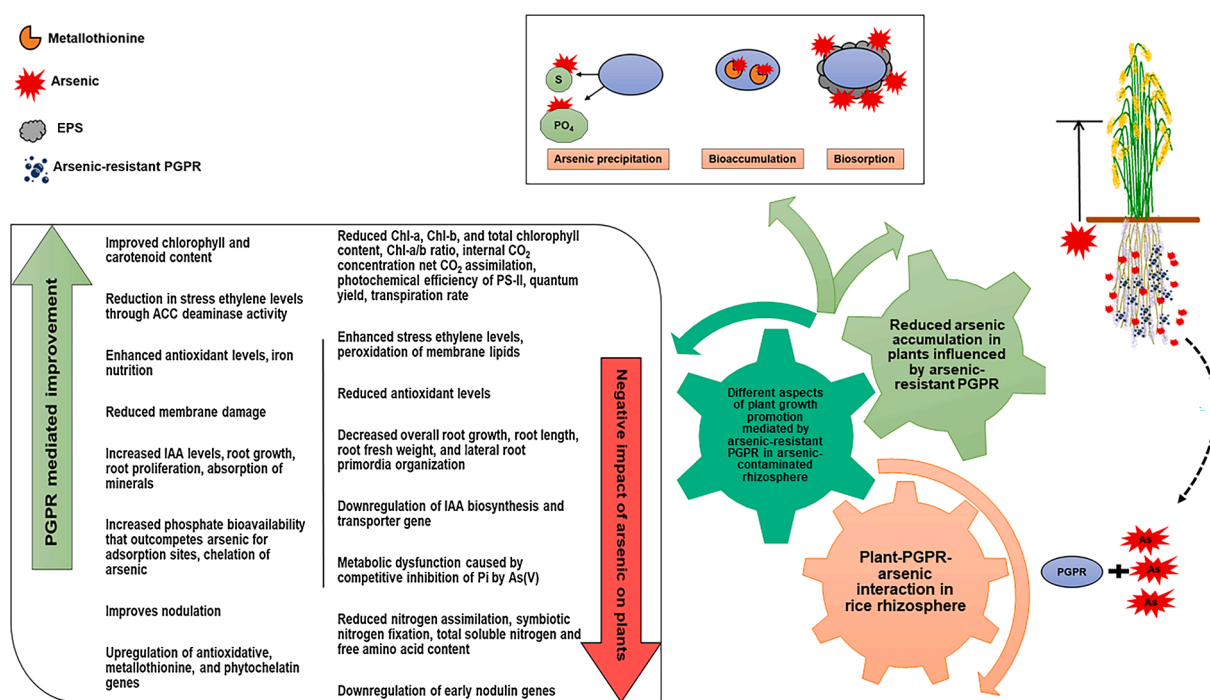


Fig. 3. Diagrammatic representation of the potential role of arsenic-resistant plant growth-promoting rhizobacteria (PGPR) in arsenic-stress alleviation in plants under arsenic-polluted soil conditions. The multipurpose role of arsenic-resistant PGPR includes assistance in nutrient acquisition (like nitrogen, phosphorus, iron, etc.), modulation of phytohormone (e.g., auxins, ethylene) and antioxidant levels (superoxide dismutase, catalase, peroxidase, glutathione reductase, etc.), reduction in membrane damage, and reclamation of photosynthetic activity. Additionally, these bacteria can also reduce the arsenic content in aerial plant parts through their different arsenic-resistance mechanisms such as exopolysaccharide mediated biosorption, bioaccumulation of arsenic through metallothionine, and arsenic precipitation promoting PGPR-mediated bioremediation.

4.1.2. Phosphorus

Phosphorus (P) is probably the second most plant growth-limiting nutrient after nitrogen. Phosphorus has a vital role in mitigating As toxicity in plants that grow in contaminated soil. As(V) utilizes phosphate transporters due to their structural similarity with phosphates (Pi) for entering into the plant system. Therefore, As(V) uptake can be strongly repressed by phosphate through competitive inhibition (Abedin et al., 2002). On the other hand, under Pi limiting condition, As(V) may restrict Pi from entering the plant by outcompeting it, which triggers Pi deficiency syndromes in plants (Finnegan and Chen, 2012). Gene expression analysis of wild-type and mutant *Arabidopsis* reveals As(V) mediated repression of Pi uptake genes. On a contrary, As(V) responsive genes were found to be induced (Catarcha et al., 2007). The addition of Pi successfully reduces As concentration in plant tissues.

Moreover, Pi addition can ameliorate As toxicity by regulating the expression of As transport, defense, and detoxification genes as reported in barley (Zvobgo et al., 2019). However, the effects of Pi on As uptake may also rely upon many other factors such as soil properties, level of As on soil, the application rate of Pi, etc. Besides, a higher proportion of As in soil requires a higher amount of Pi application which is practically not feasible due to environmental concerns (Lee et al., 2016).

In this concern, there are plentiful PGPR that not only solubilizes Pi in As-contaminated soil conditions but also provide assistance in plant growth. The metalloid adversely affects the growth of As-sensitive plants by reducing the bioavailable Pi. The deficiency can be overturned, utilizing the Pi solubilizing activity of As-resistant PGPR (Das et al., 2014; Xiao et al., 2020) (Fig. 3). The concentration of bioavailable Pi in rhizospheric soil inoculated with PGPR was found to be higher than that of control (Xiao et al., 2020). This is in turn also reduces the soil pH to a substantial level due to production of organic and inorganic acid production during the Pi solubilizing process (Das et al., 2014; Srivastava et al., 2013). Arsenic-resistant bacteria are also capable of solubilizing P from both inorganic (FePO_4 and phosphate rock) as well as organic (phytate) sources via enzymatic and/or non-enzymatic processes (Ghosh et al., 2015a). Numerous As-resistant PSB has been isolated from different sources and some of them were found to solubilize P quite efficiently. For example, multiple strains from the genera *Pseudomonas*, *Rahnella*, and *Paenibacillus* are found to solubilize P ranging between $4.1\text{--}10.5\text{ g L}^{-1}$ in As(V) stressed condition (Shagol et al., 2014). Besides, *Acinetobacter lwoffii* solubilizes $442\text{ }\mu\text{g L}^{-1}$ P in As(V) amended medium ($25\text{ }\mu\text{g L}^{-1}$); *Pseudomonas* sp. ASR1 found to solubilize $222.5\text{ }\mu\text{g L}^{-1}$ P, *Enterobacter* sp. ASR10 and *Acinetobacter* sp. solubilize $202.8\text{ }\mu\text{g L}^{-1}$ and $213.88\text{ }\mu\text{g L}^{-1}$ P respectively (Das et al., 2014; Das and Sarkar, 2018; Srivastava and Singh, 2014). Other such examples are listed in Table 2. Therefore, it can be said that the As-resistant PSB performs a dual function – it increases P uptake by plants improving its nutrition and overall growth; on the other hand, it also increases P bioavailability that may compete with As for adsorption sites that help to reduce As uptake protecting plants from As toxicity (Pinter et al., 2017).

4.1.3. Iron

Iron (Fe) is an indispensable nutrient element and is required for all forms of life. Despite being the fourth most abundant element in Earth's crust, it is hardly found in soluble form in aerobic soils. It exists mainly as ferric (Fe^{3+}) hydroxide and oxyhydroxide forms, which are largely insoluble, making it difficult for plants and microorganisms to assimilate (Glick, 2012; Rajkumar et al., 2010). In rhizospheric soil, only a limited amount of soluble iron is available by default, and all the organisms occupying that space including plants, and microbes, compete for the same (Glick, 2012). Things become more complicated in heavy metal (loid) contaminated soil. As we all know heavy metal (loid)s interfere with the developmental processes, metabolisms, and mineral nutrition of plants. Particularly, heavy metal (loid)s like As, Cd, Co, Ni, Zn, etc. mimic Fe deficiency-induced chlorosis which is an indication of disruption in Fe homeostasis (Lešková et al., 2017; Pandey and Sharma, 2002; Shaibur et al., 2009).

There are two strategies for Fe acquisition in higher plants– i) The members of Poaceae were found to secrete phytosiderophores like mugineic acid (in barley) and avenic acid (in oats) that able to chelate Fe and increases Fe uptake. However, these phytosiderophores have a lower affinity for Fe than their bacterial counterparts. Moreover, in the metal stressed condition, the efficiency of this strategy is questionable plus this strategy is exclusively for the members of Poaceae; ii) a large number of dicots and monocots (except Poaceae) do not synthesize phytosiderophores, instead, they rely upon other means for Fe acquisition such as increased Fe^{3+} reductase activity, the release of reducing agents such as phenolics, and acidification of rhizosphere. Nevertheless, the latter strategy is less efficient than the former one (Rajkumar et al., 2010). Siderophore-producing bacteria could help effectively to overcome the shortcomings of the said strategies. Plants can draw Fe from bacterial siderophores via various mechanisms such as chelate degradation and subsequent release of Fe, direct uptake of ferrisiderophore complexes, or by a ligand exchange reaction (Rajkumar et al., 2010).

Some bacteria can produce low molecular weight (400–1500 Dalton) Fe-chelating compounds are known as siderophores that have a high affinity ($K_a > 10^{30}$) towards Fe^{3+} which enables bacteria to survive in Fe-limiting environment (Glick, 2012; Kong and Glick, 2017). Siderophores are categorized into four major groups depending upon their iron-binding moieties – catecholate (e.g., enterobactin, salmochelin MGE), hydroxamate (e.g., deferroxamine/deferrioxamine B), carboxylate (e.g., staphyloferrin A), and mixed type (having elements of two or more siderophore families. e.g., yersiniabactin, pyochelin, azotobactin, pyoverdine) (Holden and Bachman, 2015) (Fig. 4). Though Fe-deficiency has been described as the primary reason for siderophore production, other factors such as soil pH and concentration of heavy metal in the external environment also determine siderophore biosynthesis (Rajkumar et al., 2010). For instance, Shagol et al. (2014) isolated 12 siderophore-producing bacterial strains belonging to the genera *Pseudomonas* and *Rahnella* from As-polluted soil. Heavy metal (loid)s may directly influence siderophore biosynthesis pathways and their regulation or it may reduce the free siderophore in the medium by forming a complex with siderophores. This halts Fe-siderophore complexation decreasing soluble Fe concentration. The Fe-deficiency then stimulates more siderophore production (Rajkumar et al., 2010). Apart from chelating Fe, bacterial siderophores are also capable of chelating other metals such as Al, Cd, Cu, Ni, Pb, Zn, Co, Mg, Sr, and As (III, V) (Dimkpa et al., 2008; Nair et al., 2007).

Several As-resistant and siderophore-producing PGP bacteria have been isolated to date by various workers. The siderophores secreted by such bacteria easily chelate insoluble Fe and make them more accessible for plants even under Fe-limiting conditions (Lakshmanan et al., 2015). This mobilization of Fe subsequently increases As solubilization and mobilization within the rhizosphere (Drewniak et al., 2008; Lakshmanan et al., 2015; Xiao et al., 2020). For example, bacterial siderophores chelate Fe^{3+} from ferric arsenates such as scorodite, and as a consequence As is also mobilized (Drewniak et al., 2008). However, bacterial siderophores are also capable of forming stable complexes with As and chelates them efficiently. On the other hand, the subsequent release of soluble Fe^{2+} by siderophore activity may promote Fe oxidizing bacteria in the rhizosphere which in turn induces Fe plaque formation that further sequesters As preventing its uptake via root (Lakshmanan et al., 2015). Examples of such As-resistant and siderophore producing PGPR include *Bacillus aryabhattai* AS6 [72 % siderophore units in 48 h in presence of As(V)], *B. nealsonii* [50.2 % siderophore units in presence of As(V)], *B. tequilensis* [42.2 % siderophore units in presence of As(III)], *Azospirillum brasilense* Cd [18.92 % siderophore units in 48 h in presence of $500\text{ }\mu\text{M}$ As(V) and 23.07 % siderophore units in 48 h in presence of $250\text{ }\mu\text{M}$ As(III)], *Pantoea* sp. strain EA106 (53.4 % siderophore units in post 24 h), *Pseudomonas* sp. PG12 and PG5 (115 and $73.2\text{ }\mu\text{M}$ deferroxamine mesylate equivalent/cell OD respectively), *B. flexus* ASO-6 ($26.8\text{ }\mu\text{M mL}^{-1}$), *Pseudomonas* sp. ASR1, ASR2 and ASR3 (32.6, 24.4, and $26.5\text{ }\mu\text{M mL}^{-1}$ respectively), *Comamonas* sp. ASR11 and ASR12 (28.3 and 30

Table 2

Properties of different arsenic-resistant PGPR and their potential role in arsenic-stress amelioration.

Arsenic-resistant PGPR	Arsenic resistance		Plant growth-promoting traits	Host Plants	Arsenic resistance mechanisms	Effect of arsenic-resistant PGPR on host plants	References
	As(III)	As(V)					
<i>Pantoea dispersa</i> AS18	3750 µg mL ⁻¹	NA	• IAA production (~90 µg mL⁻¹) • Phosphate solubilisation (~75 ppm) • ACCD activity (~0.008 nanogram α-ketobutyrate/mg protein/h) • Siderophore production • N₂ fixation • (0.34 µg N fixed/h/mg/protein) • HCN production	<i>Oryza sativa</i> L.	Bioaccumulation of As + Reduction of As(V) to As (III)	Improvement in seed germination %, relative root and shoot length, root and shoot dry weight and fresh weight; and seedling vigour index+Enhanced antioxidative enzymatic activities (SOD, CAT), proline content, amylase activity, total sugar and protein content, chlorophyll contents; reduced protease activity, MDA and stress ethylene levels+ Reduced As uptake	Ghosh et al., 2021
<i>Bacillus nealsonii</i> ARP2	NA	350 mM	• IAA production (30.6 µg mL⁻¹ for ARP2, 26.2 µg mL⁻¹ for ART2) • Phosphate solubilisation (62.3 µg mL⁻¹ for ARP2, 48.8 µg mL⁻¹ for ART2) • Siderophore production (50.2 % for ARP2, 42.2 % for ART2)	<i>Oryza sativa</i> L.	As(V) reduction activity (in ARP2) 37.5 µM min ⁻¹	Increased root length, seed germination and root biomass	Pandey et al., 2020
<i>Bacillus tequilensis</i> ART2	18 mM	NA	• EPS production (34.4 µg mL⁻¹ for ARP2, 22.4 µg mL⁻¹ for ART2) • NH₃ production (6.2 µg mL⁻¹ for ARP2, 4.2 µg mL⁻¹ for ART2)	<i>Oryza sativa</i> L.	As(III) oxidation activity (in ART2) 21.8 µM min ⁻¹	Increased root length, shoot length, weight, higher tiller no. in multiple cultivars + Reduction in As content in root and shoot	Rahman et al., 2020
<i>Bacillus</i> sp. GIS1	2 mM	300 mM	• IAA production (43 mg mL⁻¹ for GIS1, 16 mg mL⁻¹ for GIS3, 89 mg mL⁻¹ for GIS5)	<i>Oryza sativa</i> L.	As(V) reduction and As(III) oxidation (all three isolates) + Intracellular As accumulation (all three isolates)	Increased root length, shoot length, weight, higher tiller no. in multiple cultivars + Reduction in As content in root and shoot	Rahman et al., 2020
<i>Acinetobacter</i> sp. GIS3	5 mM	450 mM	• Siderophore production (hydroxamate type for GIS1, catechol type for GIS3, both type for GIS5)	<i>Oryza sativa</i> L.	Biofilm formation, EPS production + As(V) reduction and As(III) oxidation + Presence of <i>ars</i> operon consists of 5'- <i>arsR1-arsR2-arsC-acr3-arsH-3'</i> genes	NA	Vezza et al., 2020
<i>Acinetobacter</i> sp. GIS5	5 mM	500 mM	• IAA Production [~70 µg mL⁻¹ & ~60 µg mL⁻¹ for 500 µM As(V) & 250 µM As (III) respectively in 48 h] • Siderophore production [15.95 % & 23.07 95 % siderophore units for 500 µM As(V) & 250 µM As(III) respectively in 48 h] • N₂ fixation • IAA production (6.36 µg mL⁻¹ for S6, 14.95 µg mL⁻¹ for S7, and 20.02 µg mL⁻¹ for S10) • Phosphate solubilisation (1.32 mg L⁻¹ for S6, 1.30 mg L⁻¹ for S7, 1.35 mg L⁻¹ for S10) • ACCD activity (5.23, 0.87, 6.50 per mg L⁻¹ of mycoprotein producing α-ketobutyric acid by ACC catalysing the removal of one amino group per hour for S6, S7 and S10 respectively) • Siderophore production (0.26 A/Ar for S6 0.98 A/Ar for S7, 1.00 A/Ar for S10)	NA			
<i>Azospirillum brasilense</i> Cd	500 µM	5 mM	• IAA production (138.1 µg mL⁻¹) • Phosphate solubilisation (phosphate solubilisation index- 2.7)	<i>Oryza sativa</i> L.		Enhanced antioxidative enzymatic activities (SOD, POD) + Improvement in root & shoot biomass, grain yield; root length, surface area and no. of root tips	Xiao et al., 2020
<i>Pseudomonas mosselii</i> S6	NA	1425–1875 mg L ⁻¹		<i>Oryza sativa</i> L.			
<i>Bacillus thuringiensis</i> S7	NA	1425–1875 mg L ⁻¹		<i>Oryza sativa</i> L.			
<i>Bacillus</i> sp. JBS-28 S10	NA	1425–1875 mg L ⁻¹		<i>Oryza sativa</i> L.			
<i>Pseudomonas citronellolis</i>	250 µg mL ⁻¹	0.3–0.5M	• IAA production (138.1 µg mL⁻¹) • Phosphate solubilisation (phosphate solubilisation index- 2.7)	<i>Cicer arietinum</i> L.	NA	Overall improvement in root length, shoot length, shoot and root dry weight + Increase in chlorophyll content, reduced MDA levels and electrolytic leakage	Adhikary et al., 2019

(continued on next page)

Table 2 (continued)

Arsenic-resistant PGPR	Arsenic resistance		Plant growth-promoting traits	Host Plants	Arsenic resistance mechanisms	Effect of arsenic-resistant PGPR on host plants	References
	As(III)	As(V)					
<i>Acinetobacter lwoffii</i> RJB-2	50 mM	125 mM	• IAA production [325 $\mu\text{g mL}^{-1}$ at 100 mg L^{-1} As(V)] • Phosphate solubilisation [442 $\mu\text{g mL}^{-1}$ at 25 mg L^{-1} As(V)] • EPS production [4.74 $\mu\text{g mL}^{-1}$ at 100 mg L^{-1} As(V)]	<i>Vigna radiata</i>	Biofilm formation	Improved root length, shoot length, plant fresh weight and plant biomass + Increase in chlorophyll and carotenoid content, reduced MDA levels and electrolytic leakage + Decreased As accumulation	Das and Sarkar, 2018
<i>Bacillus aryabhattai</i> AS6	20 mM	100 mM	• IAA production (166 $\mu\text{g mL}^{-1}$) • Phosphate solubilisation (3 mg L^{-1}) • ACCD activity (2022 nmol α-ketobutyrate/mg protein/h) • Siderophore production (72 % siderophore units) • EPS production (44 $\mu\text{g mL}^{-1}$) • N_2 fixation (0.32 $\mu\text{g N fixed/h/mg/protein}$) • HCN and NH_3 production • IAA production (20 $\mu\text{g mL}^{-1}$ for both strains) • Siderophore production (8.0 nanomoles for AB402, 7.0 nanomoles for AB403) • EPS production	<i>Oryza sativa</i> L.	Bioaccumulation of As(V) up to 50 % and As(III) up to 31 % + Reduction of As (V) to As (III) (6.6 mM after 24 h)	Improvement in seed germination %, relative root and shoot length and seedling vigour index + Enhanced antioxidative enzymatic activities (SOD, CAT), proline content, amylase and decreased protease level	Ghosh et al., 2018
<i>Kocuriva flava</i> AB402	35 mM	NA	• IAA production (20 $\mu\text{g mL}^{-1}$ for both strains) • Siderophore production (8.0 nanomoles for AB402, 7.0 nanomoles for AB403) • EPS production	<i>Oryza sativa</i> L.	As adsorption and As accumulation within bacterial cell	Overall improvement in root length, shoot length, dry weight and wet weight + Improved chlorophyll content	Mallick et al., 2018
<i>Bacillus vietnamiensis</i> AB403	20 mM	NA			Biofilm formation		
<i>Bacillus licheniformis</i> Rt4M10	30 mM (pH 9) 20 mM (pH 7)						
<i>Micrococcus luteus</i> Rz2M10	40 mM (pH 9) 30 mM (pH 7)	NA	• IAA production • Phosphate solubilisation • Siderophore production • N_2 fixation	Grapevine cv. Malbec	NA	Increased root and shoot biomass + Increased protein (ML), APX (BL), CAT (BL and ML), POX(ML) content	Pinter et al., 2017
<i>Pseudomonas fluorescens</i> Rt6M10	5 mM						
<i>Luteimonas aestuarii</i> L22					As(V) reduction (L22, L23, L24, L25, T22, N51, N53)		
<i>Escherichia fergusonii</i> L23							
<i>Bacillus subtilis</i> L24							
<i>Psychrobacter faecalis</i> L25							
<i>Klebsiella pneumoniae</i> T22			• IAA production (all isolates except L21 and L22; highest L23 70 $\mu\text{g mL}^{-1}$)				
<i>Citrobacter freundii</i> N51							
<i>Klebsiella oxytoca</i> N53	50 mM (all isolates)	50 mM (all isolates)	• Phosphate solubilisation (L23, L25, L26, T21, T24, N51, N52) • HCN production (L21, L22, T21, T22, T23, T25, N52, N53)	<i>Triticum aestivum</i>	As(III) oxidation (L21, L26, T21, T23, T24, T25, N52) + Presence of <i>arsB</i> (T24, N52, N53) and <i>acr3</i> (L21, L23, T22, T24, T25, N53) gene	Improved root length, shoot length, no. of roots, fresh weight, and dry weight	Qamar et al., 2017
<i>Klebsiella pneumoniae</i> L21							
<i>Bacillus safensis</i> L26							
<i>Escherichia fergusonii</i> T23							
<i>Acinetobacter lwoffii</i> T24							
<i>Stenotrophomonas maltophilia</i> T25							
<i>Citrobacter freundii</i> N52							
<i>Bacillus flexus</i> ASO-6	32 mM	280 mM	• IAA production (48.2 $\mu\text{M mL}^{-1}$) • Phosphate solubilisation (93.5 $\mu\text{g mL}^{-1}$) • ACCD activity (5.2 μM α-ketobutyrate/mg protein/h) • Siderophore production (26.8 DFM mL^{-1})	<i>Oryza sativa</i> L.	As(III) oxidation	Improvement in seed germination, seedling vigour, root biomass, straw yield and grain yield + Improved chlorophyll and carotenoid content + Significant decrease in As uptake from root to shoot and As accumulation in straw and grain	Das et al., 2016
		10 mM			Biosorption	NA	

(continued on next page)

Table 2 (continued)

Arsenic-resistant PGPR	Arsenic resistance		Plant growth-promoting traits	Host Plants	Arsenic resistance mechanisms	Effect of arsenic-resistant PGPR on host plants	References
	As(III)	As(V)					
<i>Rhizobium radiobacter</i> VBCK1062			• EPS production [56 mg L ⁻¹ with 5 mM As(V)]	<i>Vigna radiata</i>			Deepika et al., 2016
<i>Exiguobacterium</i> sp. As-9	180 mM	700 mM	• IAA production [10–15 µg mL ⁻¹ with As(V)] • Phosphate solubilisation [53.4–67.7 µg mL ⁻¹ with As(V)] • EPS production [20–28 µg mL ⁻¹ with As(V)]	<i>Vigna radiata</i>	NA	Improved root length, shoot length, biomass of roots and shoots + Improved chlorophyll content + Increased SOD, CAT, APX, GR, and GPX activity + Decreased As(V) and As(III) accumulation in root and shoot	Pandey and Bhatt, 2016
<i>Brevundimonas diminuta</i> NBRI012		150 mM	• IAA production (57.83 µg mg ⁻¹) • Phosphate solubilisation (96.5 µg mL ⁻¹) • ACCD activity (13.8 % µM α-ketobutyrate mg ⁻¹ h ⁻¹) • Siderophore production	<i>Oryza sativa</i> L.	As accumulation within bacterial cell	Enhanced root length, shoot length, dry weight, fresh weight, no. of spikelet + Increased total chlorophyll, carotenoid, anthocyanin, proline, and protein content, reduced MDA level + Reduced As accumulation in shoots	Singh et al., 2016

ACCD- 1-aminocyclopropane-1-carboxylate deaminase; APX- ascorbate peroxidase; As- Arsenic; As(III)- arsenite; As(V)- arsenate; CAT-catalase; EPS- exopolysaccharide; GPX- glutathione peroxidase; GR- glutathione reductase; HCN- hydrogen cyanide; IAA- indole-3-acetic acid; MDA- malonaldehyde; N₂- nitrogen; NH₃- ammonia; PGPR- plant growth promoting rhizobacteria; POX- peroxidase; SOD- superoxide dismutase.

µM mL⁻¹ respectively), *Kocuria flava* AB402 and *B. vietnamiensis* AB403 (8 and 7 nanomoles of siderophores after 48 h of incubation with or without As respectively) (Das et al., 2014; Ghosh et al., 2015a; Lakshmanan et al., 2015; Ghosh et al., 2018; Mallick et al., 2018; Pandey et al., 2020; Vezza et al., 2020). Surprisingly, in many cases, siderophore production remains unaltered even in the presence of toxic heavy metal (loid) such as As (Mallick et al., 2018; Vezza et al., 2020). The As-resistant PGPR is known to improve Fe nutrition, enhances plant biomass, influences chlorophyll synthesis and root proliferation in plants like *Brassica juncea* and *Cicer arietinum* (Srivastava et al., 2013; Srivastava and Singh, 2014). The presence of such PGPR increased plant biomass up to 45 % and increased As removal efficiency up to 35 % in presence of mixed inoculum containing *Pseudomonas* sp., *Delftia* sp., *Bacillus* sp., *Variovorax* sp., and *Pseudoxanthomonas* sp. (Lampis et al., 2015). Two particular genes in rice OsIRT1 and OsYSL15 expressed in roots encodes specific Fe transporters that are induced by Fe-limiting conditions at least in early germination stages (Ishimaru et al., 2006; Inoue et al., 2009; Lee and An, 2009; Lee et al., 2009). Inoculation with *Pantoea* sp. strain EA106 downregulates both OsIRT1 and OsYSL15 expression which is suggestive of Fe abundance in the rhizosphere mobilized predominantly by *Pantoea* sp. strain EA106 under As-rich environment (Lakshmanan et al., 2015). The increase in soluble Fe in the rhizospheric soil of *Pteris vittata* was also found to be linked with pyochelin type of siderophore produced by As-resistant bacteria like *Pseudomonas* sp., *Comamonas* sp., and *Stenotrophomonas* sp. (Ghosh et al., 2011). Siderophore-producing bacteria also showed greater As(III) oxidation and As(III) resistance this is since several As(III) oxidase enzymes require Fe for their activation (Drewniak et al., 2008; Ghosh et al., 2015b). Siderophore-producing As-resistant PGP bacteria affects plant growth and development by i) its siderophore producing ability that indirectly prevents heavy metal(loid) entry into plant, ii) facilitating plants in Fe acquisition under Fe-limiting conditions, and iii) promoting plant growth through its other PGP traits.

4.2. Modulation of phytohormones

4.2.1. Auxins

Auxins are phytohormones that are found to be involved in all major plant physiological processes including root initiation and development. Among different kinds of naturally occurring auxins, indole-3-acetic

acid (IAA) is by far the most common and the most studied one. Starting from root apical growth, IAA is involved in cell enlargement, cell division, vascular tissue differentiation, tropistic responses, apical dominance, leaf senescence, leaf and fruit abscission, fruit setting, and growth, assimilate partitioning, fruit ripening, flowering, growth of floral parts, and promotion of femaleness in dioecious flowers (Davies, 2010). However, some microorganisms are also known to produce IAA and a major point of interest during plant-microbe interaction in the rhizosphere. Reports suggest almost 80 % of total rhizospheric bacteria are capable of producing IAA (Spaepen and Vanderleyden, 2011). The bacterial IAA might with the plants' endogenous IAA pool altering the overall IAA level in plants (Fig. 5).

Arsenic negatively affects IAA biosynthetic (OsASA2 and OsYUCCA2) and transporter gene (AUX1 and PIN5b) expression disrupting IAA biosynthesis, transport, and localization (Ronzan et al., 2019). Similar results were also obtained from experiments on *Arabidopsis*, in which Cd and As, alone or together affects post-embryonic root formation and development by tweaking root quiescent center, altering expression of YUC6 (involved tryptophan dependent IAA synthesis) and modulating auxin efflux (PIN1), and influx carriers (LAX3) (Fattorini et al., 2017). The endogenous levels of IAA and indole-3-butyric acid (IBA) are also found to be declined upon As treatment in *Brassica juncea* (Srivastava et al., 2013). On the other hand, external application of IAA proved to increase phytoremediation efficacy of plants like sorghum and sunflower whereas total biomass and antioxidative enzymatic activity [superoxide dismutase (SOD), catalase (CAT), and peroxidase (POX)] significantly increased in *Pteris ensiformis* and *P. cretica* var. *nervosa* that were exposed to As(V), As(III), and DMA (Lyubun, 2009; He et al., 2017).

In this concern, many PGPR strains possess IAA producing trait that enables them to sustain IAA production even in As-rich environments that allows growth promotion of the plants that grows under such conditions. The examples include *Brevundimonas intermedia* B10 (IAA production 52.45 mg L⁻¹), *Bacillus aryabhatai* AS6 (166 µg mL⁻¹), *Pseudomonas citreolensis* KM594397 (138.1 µg mL⁻¹), *Staphylococcus arlettae* NBRIEAG-6 (41.07 µg mg⁻¹), *Acinetobacter* sp. (60.93 µg mg⁻¹), *Brevundimonas diminuta* NBRI012 (57.83 µg mg⁻¹), *Rahnella aquatilis* JS119 (65.3 µg mL⁻¹), *Rahnella aquatilis* JS120 (57.2 µg mL⁻¹), *Acinetobacter* sp. GIS5 (89 mg mL⁻¹), *Bacillus* sp. GIS1 (43 mg mL⁻¹), *Escherichia fergusonii* L23 (70 µg mL⁻¹), *Klebsiella oxytoca* N53 (47 µg

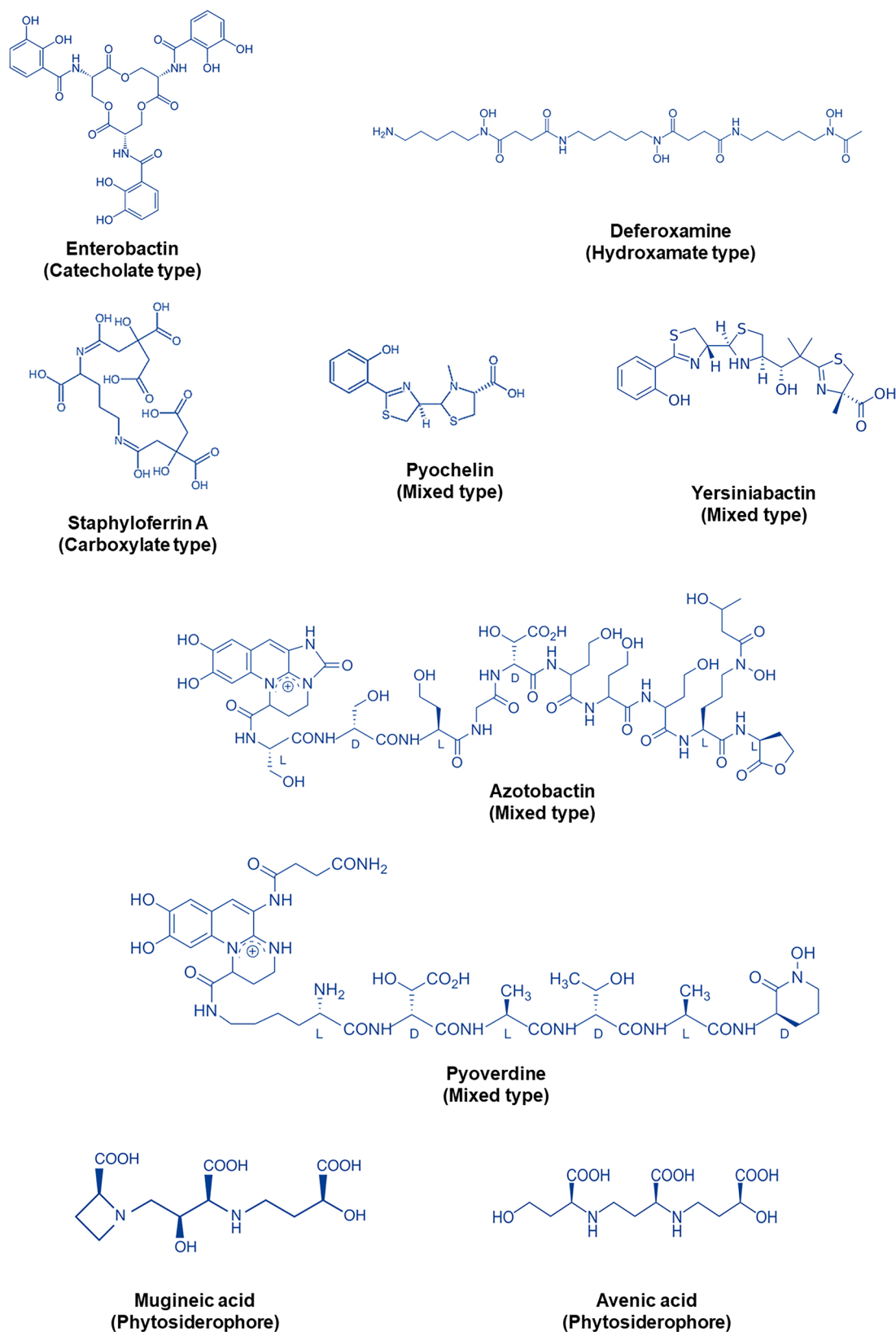


Fig. 4. Chemical structures of different types of microbially-produced siderophores and phytosiderophores.

mL⁻¹), *Bacillus flexus* ASO-6 (48.2 μ M mL⁻¹), *Pseudomonas* sp. ASR1 (58.4 μ M mL⁻¹), and *Pantoea dispersa* (~90 μ g mL⁻¹) to name a few (Adhikary et al., 2019; Das et al., 2014, 2016; Ghosh et al., 2018, 2021; Qamar et al., 2017; Rahman et al., 2020; Shagol et al., 2014; Singh et al.,

2016; Srivastava et al., 2013; Srivastava and Singh, 2014; Soto et al., 2019). It is also reported in some cases that IAA production is not affected by high concentrations of the metalloid, as found in *A. brasilense* Cd, where exposure to 500 μ M As(V) and 25 μ M As(III) showed no

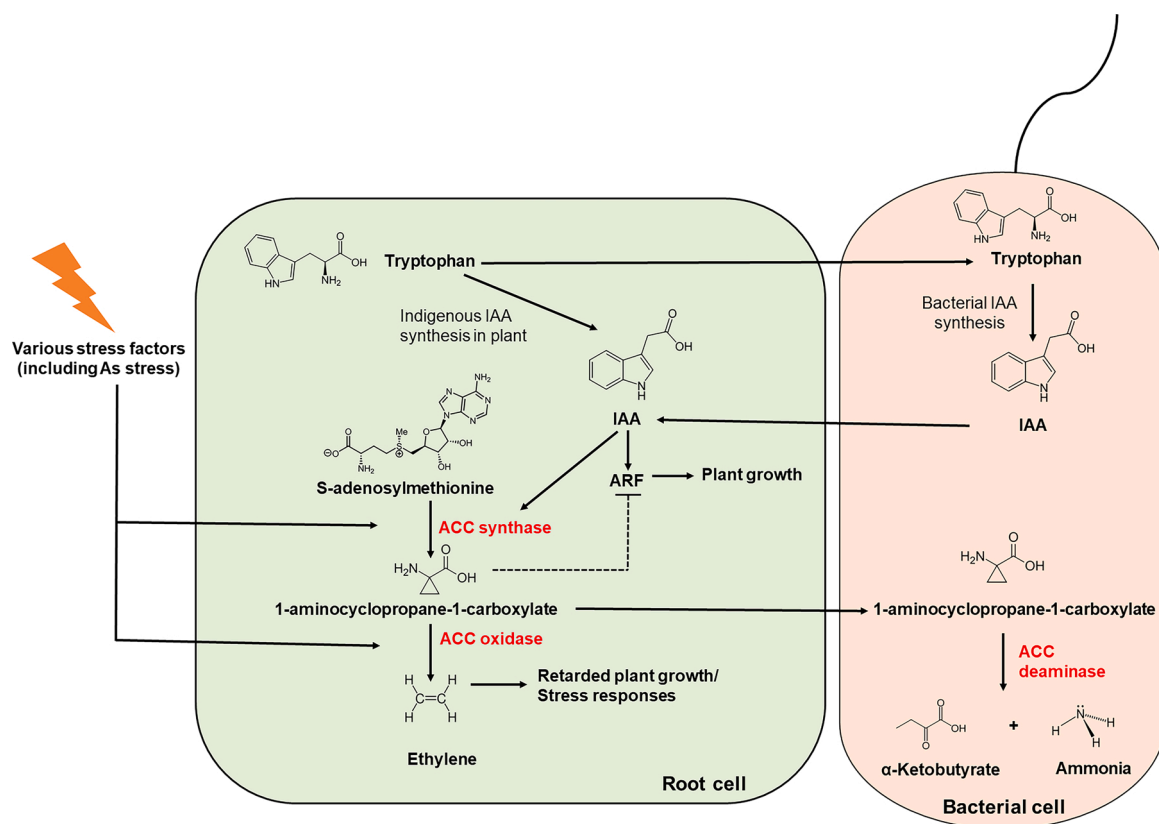


Fig. 5. Schematic depiction of how arsenic-resistant PGPR that produces both indole-3-acetic acid (IAA) and 1-aminocyclopropane-1-carboxylate deaminase (ACCD) enhances plant growth under As-contaminated soil conditions. PGPR may receive the precursor tryptophan from plants that stimulate their IAA production ability. Microbially-produced IAA further increases plants' indigenous IAA pool that subsequently upregulates auxin responsive factors (ARF) resulting in plant growth. S-adenosylmethionine is converted to ACC by the enzyme ACC synthase; ACC is converted to ethylene by ACC oxidase. Under stressed conditions, ethylene production increases as a stress-responsive mechanism. Besides, IAA stimulates ACC synthase gene expression elevating ethylene levels further. Rising ethylene levels blocks transcription of ARF. Additionally, some amount of ACC made inside plants is released from root cells, taken up by PGPR which is then converted into NH_3 and α -ketobutyrate through ACCD activity of PGPR.

significant difference in IAA production ($\sim 70 \mu\text{g mL}^{-1}$) with respect to control and a little decrease in IAA content ($\sim 60 \mu\text{g mL}^{-1}$) only observed under most toxic environment [$250 \mu\text{M As(III)}$]. However, the decline in IAA production is directly related to a decline in bacterial growth and viability (Vezza et al., 2020). Similar findings were also proposed by Mallick et al. (2018). In addition, they also found that PGPR like *Kocuria flava* and *Bacillus vietnamiensis* produce IAA even in the absence of tryptophan and As with the latter one showing better efficacy (Mallick et al., 2018). Conversely, the viable cell-count, as well as IAA production increased with increasing As(V) concentration [IAA production increased from $204 \mu\text{g mL}^{-1}$ at control to $325 \mu\text{g mL}^{-1}$ at 100 mg L^{-1} of As(V) in the media] in *Acinetobacter lwoffii* (Das and Sarkar, 2018). It is also assumed that microorganisms are so adapted to stressful conditions that even the presence of As in the medium does not affect IAA production (Soto et al., 2019). The IAA released by PGPR enhances plant growth by stimulating elongation of plant cells or increasing cell division thereby expanding total root surface area enhancing mineral nutrient uptake from soil and not only that it also increases As absorption via roots indicating its indirect role in As mobilization and accumulation (Srivastava and Singh, 2014; Soto et al., 2019; Vezza et al., 2020; Wang et al., 2011). Inoculation with such As-resistant IAA producing strains successfully augmented growth and development in plants like *Oryza sativa*, *Cicer arietinum*, *Brassica juncea*, *Vigna radiata*, etc. (Srivastava et al., 2013; Srivastava and Singh, 2014; Das and Sarkar, 2018; Ghosh et al., 2018; Vezza et al., 2020; Xiao et al., 2020). Bacterial IAA has also been reported to enhance the survival of more As-sensitive rhizobacterial strains that exist in close proximity as found in the case of *Azospirillum brasilense* Az39 that promotes survival of *Bradyrhizobium*

japonicum E109 under As-stressed condition (Armendariz et al., 2019). A cross-talk between IAA and ACCD activity in As-resistant PGPR ameliorates inhibitory effects of ethylene and further promote plant growth in As-contaminated soil (Das et al., 2016).

4.2.2. Ethylene

It is a widely known fact that the gaseous plant hormone, ethylene is overproduced as “stress ethylene” in response to various environmental stresses (Glick et al., 2007; Saraf et al., 2010; Soni et al., 2018). The production of this “stress ethylene” is detrimental to normal plant functioning.

In all likelihood, arsenic provokes “stress ethylene” biosynthesis in plants and as a result, ACC accumulates within the roots (Srivastava and Singh, 2014). The production of “stress ethylene” further exaggerates the detrimental effects of several environmental stresses. Arsenic-induced “stress ethylene” production is one of the main reasons behind the inhibition of root growth and stunted plant growth (Ghosh et al., 2018). Concerning plant growth and development in As-spiked soil, ACCD producing As-resistant PGPR provides a much-needed impetus by reducing “stress ethylene” levels invoked primarily by As. In As-treated seedlings, an elevated level of “stress ethylene” production was recorded (Ghosh et al., 2018). However, the surge of ethylene was found to be checked with the application of ACCD producing As-resistant PGPR. Not only that, the reduction in ethylene levels was found comparable to the chemical ethylene inhibitor aminoethoxyvinyl glycine (AVG) (Ghosh et al., 2018). The chemical ethylene inhibitors like AVG, 1-methylcyclopropane (1-MCP) and silver thiosulfate, etc. are currently being used widely in crop practices and is recommended be

replaced by more environmentally friendly option such as ACCD producing PGPR (Soni et al., 2018). Moreover, inoculation with ACCD producing As-resistant PGPR such as *P. mosselii*, *Bacillus* sp. JBS-28, *B. aryabhattai*, *Acinetobacter* sp. nbri05, *Staphylococcus arlettae* NBRIEG-6, *Brevundimonas diminuta*, *B. flexus* ASO-6, *Pantoea dispersa*, etc. found to improve overall plant growth including increasing grain yield, straw biomass, and root length in crops like *Oryza sativa* L., *Cicer aritenum*, *Brassica juncea* (L.) Czern. Var. R-46 and helps them to thrive in As amended soil countering the adverse effects of “stress ethylene” (Das et al., 2016; Ghosh et al., 2018, 2021; Singh et al., 2016; Srivastava et al., 2013; Srivastava and Singh, 2014; Xiao et al., 2020). The reason behind such an impactful effect of PGPR lies in the fact that microbes can potentially interfere with all the regulatory steps of ethylene biosynthesis in plants (Ravanbakhsh et al., 2018). Besides, ACCD producing bacteria effectively down-regulate genes that are involved in ethylene-induced plant stress responses in root while up-regulating genes that are meant for plant growth such as cell division and proliferation (Hontzeas et al., 2004). Arsenic-resistant PGPR that capable of producing both IAA and ACCD can significantly lower stress ethylene levels in plants grown in As-contaminated soil conditions due to an underlying cross-talk between IAA and ACCD. Such PGPR may fetch IAA-precursor tryptophan in the form of root exudate. Microbially-produced IAA coupled with plants' indigenous IAA pool promotes plant growth by regulating auxin responsive factors (ARF). Ethylene production is increased under stressed condition (e.g., As stress) as a stress-responsive mechanism. Further, elevated IAA levels can also stimulate ethylene production which then negatively regulates ARF expression. Moreover, the ACC made during this period can be converted into NH_3 and α -ketobutyrate by ACCD activity of the PGPR (Glick, 2014; Gamalero and Glick, 2015) (Fig. 5).

4.3. Regulation of antioxidative defense system in plants

Plant when exposed to As produces ROS that induces oxidative stress leading to cellular damage. As-induced overproduction of ROS can be regulated through antioxidant enzymes as they tend to scavenge free radicals. Higher levels of H_2O_2 have been observed with increasing concentration of As in horseradish [highest at 45 and 60 $\mu\text{g L}^{-1}$ of As(V)] and *Vicia faba* [highest at 20 μM As(III)] (Ahmad et al., 2020; Kofronová et al., 2019). Similarly, in response to As(III) increased H_2O_2 and O_2^- were observed in leaves of ryegrass and rice (Jung et al., 2019; Li et al., 2019). To counter the toxic free radicals modulation of various enzymatic and non-enzymatic antioxidants such as SOD, ascorbate peroxidase (APX), glutathione peroxidase (GPX), glutathione reductase (GR), CAT, guaiacol peroxidase (GOPX), glutathione S-transferase (GST), dehydroascorbate reductase (DHAR), monodehydroascorbate reductase (MDHAR), reduced glutathione (GSH), oxidized glutathione (GSSG), and proline has been reported (Ahmad et al., 2020; Begum et al., 2016; Gupta et al., 2009; Gururani et al., 2013; Kofronová et al., 2019; Shri et al., 2009). However, differential activities of antioxidants were evident upon As exposure. For example, in rice, SOD, CAT, and APX up-regulated in both roots and leaves in response to As(III). Conversely, the ascorbate (AsA)/DHA and GSH/GSSG ratio were found to be decreased along with MDHAR, DHAR, and GR (Jung et al., 2019). Several other studies also reveal otherwise. A study reported by Ahmad et al. (2020) showed increased SOD, CAT, APX, GR, GSH, and GSSG activities upon As(III) treatment whereas DHAR and MDHAR levels were said to be decreased. In another study, CAT levels were down while APX, SOD, and POX levels were up (Srivastava et al., 2017). In general, it can be said that mild doses of As may trigger antioxidant activity significantly up to a level but as toxicity levels piled up, the activities were found to be declined (Farooq et al., 2016; Gupta et al., 2009; Kofronová et al., 2019). However, this is not true for all cases. The antioxidative defense system works differently in various plant species under As stress and their modulation is perhaps dose-dependent. The statement is further justified by the fact that the peak of the enzymatic activity for

different antioxidant enzymes is found at different As concentrations (Kofronová et al., 2019).

Microorganisms ameliorate oxidative stress induced by metals largely by – reducing free radical formation, activating various antioxidants and antioxidative enzymes, modulating ROS-scavenging pathways to main ROS homeostasis, and recycling ascorbate-glutathione enzymes to elevate the redox state of the cell (Khanna et al., 2019). The microbial interference in the plant's antioxidative defense machinery is also evident in the case of As toxicity. The effect of microbes on plant antioxidative defense varies greatly and it seemed to be species and/or strain-specific and in some instances, it is antioxidative enzyme-specific. Other factors such as metalloid concentration in soil and the effect of other plant growth-promoting traits of PGPR may also play a decisive role in such interaction. However, the outcome of such bacterial influence always proved to be beneficial for plants in some way or other as documented by several researchers. In *Vigna radiata*, As(III) and As(V) treatments (100 mg kg^{-1} soil) significantly increased SOD, APX, GR, CAT, and GPX activity (66%–97%) in both root and shoot. The effects were found more pronounced in more toxic As(III) treatments. However, when seedlings were grown in presence of bioinoculant *Exiguobacterium*, the enzymatic activities were reduced and the values were found comparable to that of control sets (without stress and bioinoculant). This result is indicative of bacteria-mediated reduction of oxidative damage due to attenuation of ROS production (Pandey and Bhatt, 2016). SOD activity in rice also showed a similar pattern when exposed to As(V) (50 $\mu\text{g mL}^{-1}$), under influence of bacteria belongs to *Bacillus cereus* cluster (Pandey et al., 2013). In some cases, SOD and CAT activities decreased with increasing As(V) concentration in soils as reported in *Populus deltoids*. Inoculation with *Agrobacterium radiobacter* significantly improved SOD activity in roots and leaves by 31 % and 51 % respectively at 150 mg kg^{-1} As amendment and by 11 % and 25 % respectively at 300 mg kg^{-1} As amendment. The activity of CAT in roots and leaves improved by 98 % and 142 % respectively at 150 mg kg^{-1} As amendment; 25 % and 32 % respectively at 3000 mg kg^{-1} As amendment (Wang et al., 2011). APX activity in grapevine up-regulated by *Bacillus licheniformis* in As(III) amended condition (150 μM). A similar trend was also observed in case of POX where enzymatic activity only increased upon bacterization with *Micrococcus luteus*. However, CAT activity increased in presence of As(III) irrespective of bacterial strain (Pinter et al., 2017). The set of results hinting about As-induced downregulation of antioxidative enzymes resulting in oxidative damage. PGPR facilitates partial procurement of antioxidative enzymatic activity of plants to combat oxidative damage imposed by As. A recent study by Ghosh et al. (2018) revealed that As(V) treatment enhanced SOD and CAT activities in rice seedlings which were further increased by 27.27 % and 62.26 % respectively by the application of an As-resistant strain *Bacillus aryabhattai*. The strain also increased proline content in a similar pattern. Identical results were also obtained by Xiao et al. (2020) in which SOD and POX activity in rice roots was promoted by three different PGPR inoculation viz. *Pseudomonas mosselii*, *Bacillus thuringiensis*, and *Bacillus* sp. JBS-28. This sort of result shows the synergistic effect of PGPR on antioxidant enzyme activity of plants that helps in reducing As-induced oxidative damage by enhancing ROS scavenging mechanisms. Therefore, the enhancement of antioxidative enzymatic activity should also be considered as a plant growth-promoting trait (Xiao et al., 2020).

A recent study on wheat revealed that PGPR such as *Brevundimonas intermedia* and *Pseudomonas gessardii* induces relative expression of plant SOD and APX gene during plant-As interaction (Soto et al., 2019). Nonetheless, the detailed mechanism of this type of interaction between PGPR and the plant's antioxidative defense machinery during As stress is still largely unknown.

4.4. Protection against membrane damage

As induces oxidative stress through ROS production which in turn may induce peroxidation of polyunsaturated fatty acid (PUFA) in the

membranes that leads to the formation of malonaldehyde (MDA) (terminal product of lipid peroxidation) (Srivastava and Singh, 2014). A high level of MDA in the As exposed plant is an indication of free radical formation in the cells and it is frequently used as an index of lipid peroxidation (Fig. 3). Plant cell membranes are primary targets of free radicals and cause membrane injury that leads to membrane destabilization (Singh et al., 2016). MDA content in *Vigna radiata* was found to be increased by 50.2 % and 82.8 % in shoots upon As(V) and As(III) treatment respectively. Whereas it was found to be increased by 42.85 % and 84.52 % in roots upon As(V) and As(III) treatment respectively (Pandey and Bhatt, 2016). As induced MDA production has also been reported in other plants like rice, *Brassica napus*, chickpea, *Vigna mungo*, etc. by various authors (Adhikary et al., 2019; Ahmad et al., 2020; Farooq et al., 2016; Jung et al., 2019; Singh et al., 2016). Now, when As treated plants were inoculated with As-resistant PGPR, MDA content was found to be reduced which is an indication of the reduction in membrane damage. For example, inoculation with *Brevundimonas diminuta* showed reduction in MDA content in rice plants treated with 10 mg kg⁻¹ and 50 mg kg⁻¹ of As(V) (Singh et al., 2016). Similarly, *Exiguobacterium* reduced MDA levels in the range of 7–37 % and *Acinetobacter lwoffii* reduced MDA in the range of 12–19 % in *Vigna radiata* (Das and Sarkar, 2018; Pandey and Bhatt, 2016). Electrolyte leakage (EL) can serve as a potent indicator of the membrane damage caused by lipid peroxidation. Higher EL values are indicative of higher membrane injury. Upon As treatment EL was increased by 107.34 % (in 10 µM As concentration) and 303.72 % (in 20 µM As concentration) in *Vicia fava* (Ahmad et al., 2020). The EL of As(V) treated *Vigna radiata* leaf tissues were found to be way higher (74.8 %) than the control (30.8 %) and is subsequently reduced to 54.9 % upon bacterial inoculation (*Acinetobacter lwoffii*) (Das and Sarkar, 2018). In chickpea too, EL was found to increase by ≥ 50 % at various As(V) concentrations. However, a significant decrease in EL was observed after applying *Pseudomonas citronellois* (Adhikary et al., 2019).

4.5. Reclamation of photosynthetic activity

Photosynthesis is probably the most important metabolism for plants as well as for the rest of the living world. Heavy metal(loid) induced impairment of photosynthesis has been reported repeatedly by various workers (Fig. 3). As affects multiple aspects of photosynthesis and as a result net photosynthesis of plants got reduced alarmingly. The reduction in photosynthesis rate upon As exposure is believed to be one of the most harmful effects caused by the metalloid (Gusman et al., 2013). The deleterious effect of As includes distortion of chloroplast ultrastructure, inhibition of chlorophyll biosynthesis pathway, and chlorophyll degradation that ultimately inhibits photosynthesis to a certain level (Chandrakar et al., 2018; Farnese et al., 2017). Reduction in chlorophyll-a, chlorophyll-b, chlorophyll- a/b ratio, and total chlorophyll content was previously reported in many plants like wheat, rice, chickpea, maize, and mung bean upon As exposure (Adhikary et al., 2019; Ahmad et al., 2020; Anjum et al., 2017; Das and Sarkar, 2018; Maglovski et al., 2019). Besides the metalloid was also found to down-regulate stomatal conductance, internal CO₂ concentration, net CO₂ assimilation, photochemical efficiency of PS-II, quantum yield of CO₂ assimilation, transpiration, etc. (Farnese et al., 2017; Farooq et al., 2016; Gusman et al., 2013). The decreased photosynthetic pigment level reflects a lack of adaptive acclimatization strategies of pigment biosynthesis in plants at high As concentrations (Singh et al., 2016). These photosynthetic pigments control the photochemistry of photosynthesis and are directly involved in carbohydrate synthesis. Therefore, the As-induced reduction of the pigments can hamper the growth and yield of the plants (Maglovski et al., 2019; Zaid et al., 2020). The decrease in pigment content might be due to altered RuBisCo activity and/or chlorophyll biosynthesis activity (particularly δ-aminolevulinic acid dehydratase and protochlorophyllide reductase) (Chandrakar et al., 2018; Yadav et al., 2016). The reaction of As with functional –SH group of

δ-aminolevulinic acid dehydratase and other Fe-containing components of chlorophyll precursors further explain the mechanism (Maglovski et al., 2019). Arsenic has also been shown to replace the central Mg atom of the chlorophyll molecule (Yadav et al., 2016). Mishra et al. (2016) extensively studied As-induced inhibition of chlorophyll biosynthesis on *Ceratophyllum demersum*, and they observed a strong decrease in the levels of most precursors of chlorophyll biosynthesis pathway even at low As doses (0.5 µM). Coproporphyrinogen-III, an early precursor in the tetrapyrrole pathway (which later converted to protoporphyrin IX and then onto Mg-protoporphyrin) was found in negligible quantity during their experiment (Mishra et al., 2016). The toxic effect of As on liquid phase photosynthetic attributes such as transpiration, stomatal conductance, chl-a fluorescence has been well documented but information related to the physiological response of plants exposed to As is scanty (Farooq et al., 2016; Gusman et al., 2013). In addition, the metalloid can also affect the CO₂ fixation process through the manipulation of Calvin cycle enzymes such as RuBisCo (Chandrakar et al., 2016; Gusman et al., 2013). The downregulation of RuBisCo (large subunit) and chloroplast 29 kDa ribonucleoproteins have also been reported previously upon As exposure (Ahsan et al., 2010). As severely affects membranes through lipid peroxidation as a result of ROS overproduction. Chloroplast is a major site for ROS production in plants and is more sensitive towards As imposed anomalies. As bring about structural changes in outer and inner chloroplast membranes. As also promotes degradation of chloroplast evidenced from the occurrence of vesiculation at the outer membrane of the organelle. As induced chlorophyll degradation and deformation of its internal membrane are said to be the major reason behind reduced pigment levels and carbon assimilation (Farnese et al., 2017). Reduced pigment level is the probable reason behind the reduced efficiency of PS-II since As was not found to disintegrate PS-II protein components (Farnese et al., 2017). The application of As-resistant PGPR has proved to nullify the toxic effects of As on photosynthetic machinery. In general, the bacterial inoculation was found to improve chlorophyll content in respective host plants and some cases, almost to the levels of control (without As-stress, and bacteria) reversing the loss caused by the metalloid. Examples include *Staphylococcus arlettae*, *Acinetobacter* sp., *Exiguobacterium*, *Bacillus flexus*, *Acinetobacter lwoffii*, *Kocuria flava*, *Bacillus vietnamensis*, *Pseudomonas citronellolis*, and *Pantoea dispersa*. (Adhikary et al., 2019; Das et al., 2016; Das and Sarkar, 2018; Ghosh et al., 2021; Mallick et al., 2018; Pandey and Bhatt, 2016; Srivastava et al., 2013; Srivastava and Singh, 2014). Many of these bacteria were found to produce siderophore and therefore speculated to improve the Fe nutrition which in turn improves chlorophyll content (Srivastava et al., 2013; Das and Sarkar, 2018). Additionally, the carotenoid content of the host plants was also enhanced in some cases using As-resistant PGPR like *Bacillus flexus* and *Acinetobacter lwoffii* that further minimizes photochemical damage (Das et al., 2016; Das and Sarkar, 2018).

4.6. Reduced arsenic accumulation within plants

The process of biosorption and bioaccumulation has been well-established as an As-resistance mechanism in bacteria. These processes not only protect bacterial cell from the acute toxicity of the metalloid but also provides a level of protection to the host plants (Fig. 3). In addition, biofilm formation is also observed in some As-resistant PGPR. Biofilm formation is assumed to be a protective mechanism against metal(loid) toxicity. EPS is the main constituent of the biofilm. Biofilm formation was found to increase at inhibitory As(III) concentrations in *Bradyrhizobium japonicum* E109 and *Azospirillum brasilense* Az39 (Armendariz et al., 2015). and corroborated with increased EPS production. Similar results were obtained in case of *Acinetobacter lwoffii* and *Exiguobacterium*, where increasing As(V) concentration (0–100 mg L⁻¹ and 400–600 mg L⁻¹ respectively) enhanced the biofilm-forming ability of the strain (Das and Sarkar, 2018; Pandey and Bhatt, 2016). The biofilm formation in *Azospirillum brasilense* Cd was found to be uninterrupted at

an As concentration that does not affect bacterial growth (Vezza et al., 2020). A significant amount of the metalloid was also detected on the EPS of such PGPR as in *Rhizobium radiobacter* and *Bacillus* sp. (Deepika et al., 2016; Rahman et al., 2020). The EPS-mediated biosorption quantified in *Halomonas* sp. is 43 mg kg⁻¹ (Mukherjee et al., 2019). A decent amount of As reported to be accumulated within the bacterial cell e.g. *Bacillus* sp. (848.33 µg g⁻¹ dry cell weight) *Halomonas* sp. (252.4 mg L⁻¹ As) (Mukherjee et al., 2019; Rahman et al., 2020). The mechanism of biosorption and bioaccumulation is also reported from several other As-resistant PGPR (e.g., *Bacillus aryabhattai*, *B. nealsonii*, *B. tequilensis*, *Brevundimonas diminuta*, *Pantoea dispersa*) which indicates that the existence of these protective mechanisms is very common among these group of bacteria (Ghosh et al., 2018, 2021; Pandey et al., 2020; Singh et al., 2016). Moreover, in all cases, improved growth of the respective host plants under metalloid toxicity can be observed. Therefore, it can be assumed that biosorption and bioaccumulation must be decisive factors in reducing As concentration in plants and thereby promoting plant growth when combined with other PGP traits.

5. Conclusion and future research challenges

PGPR are unique microflora and active colonizers around crop root systems. Their successful colonization is useful in several ways starting from plant growth promotion to metal(loids) stress alleviation. From a detailed discussion in this review literature, it is clear that the role of PGPR in As-stress alleviation in plants under contaminated soil ecosystem is multifaceted. The detailed mechanism of many of these As-stress alleviating mechanism that takes place during plant-PGPR interaction is still unknown and the future beholds several research challenges including (i) in-depth analysis of the biochemistry behind the synergistic interaction between As stress laden plant and its root colonizing As-resistant PGPR; (ii) identification of the genes that are expressed during plant-PGPR interaction under As-contaminated soil; (iii) isolation of native As-resistant PGPR and its successful optimization for potential use in sustainable agriculture; (iv) exploration of the full potential of As-resistant PGPR and other metalloids resistant microorganisms such as fungi and study the effect of their co-inoculation as the soil microbiota contains a variety of microorganisms including bacteria and fungi; (v) effect of other abiotic stresses on plant-PGPR interrelation under As-polluted environment; (vi) study the effect of As on plants as well as on PGPR when combined with other heavy metal(loids) and how does multiple heavy metal(loids) affect plant-PGPR relationship since most meta(loid)-stressed crop fields are contaminated with multi metal (loid)s; (vii) extensive study of the As resistance mechanisms and genes responsible for those mechanisms in PGPR; (viii) metagenomic study of microbial consortia in the metal(loid) polluted site which can improve the strategies for assessing the impact of metal(loid) on the ecosystem and for cleaning up environments using metal(loid)-resistant PGPR; (ix) extensive field trials of such potent As-resistant PGPR is required since results obtained in green house studies and under *in vitro* conditions may differ significantly.

In a nutshell, it can be depicted that As-resistant PGPR possesses an immense potential that enables them to ameliorate metalloid toxicity from plants that grow in the contaminated sites. This bioremediation capability of such PGPR can be further utilized in developing sustainable agriculture. However, some fundamental questions still need to be answered. For example, what is the fate of the metalloid that accumulated within the bacterial cell? Why some As-resistant PGPR converts less-toxic forms of As to relatively more toxic forms? and how does that affect the host plant? Nonetheless, a better understanding of all these aspects will further decipher the enigma of plant-microbe interaction in As-polluted agricultural fields.

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Author contribution statement

SM, KP, and TKM conceptualized the theme of the review. SM wrote the manuscript. SKG, PP, TM, and TS helped in the data curation of Tables. KP and TKM revised and finalized the manuscript. TKM supervised all authors. All authors finally read and approved the manuscript.

Declaration of Competing Interest

The authors report no declarations of interest.

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